Haloalkanes and Haloarenes

Question1

Consider all the structural isomers with molecular formula $C_5H_{11}Br$ are separately treated with $KOH(aq)$ to give respective substitution products, without any rearrangement. The number of products which can exhibit optical isomerism from these is ____.

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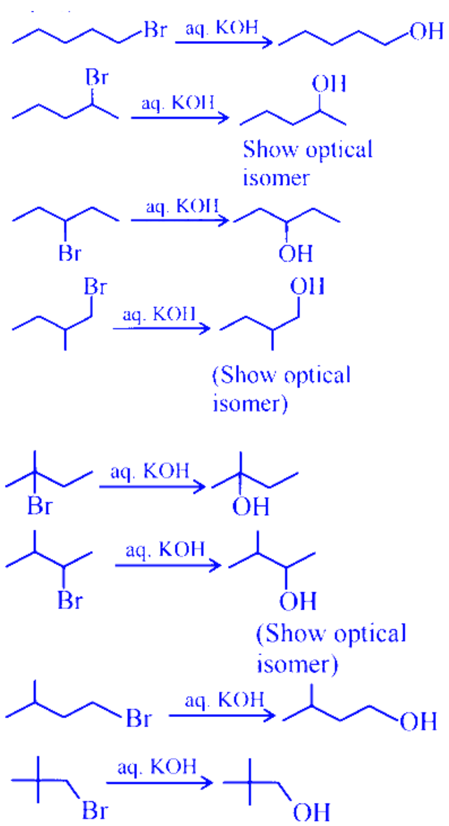
Answer: 3

Solution:



As per the language given and considering the condition we are going with answer 3 and considering both active isomers we will be giving 6 too.

Question2

RMgI when treated with ice cold water liberated a gas which occupied $1.4 dm^3/g$ at STP. The gas produced is further reacted with iodine in presence of HIO_3 to give compound (X). Compound (X) in presence of Na and dry ether produced compound (Y). Molar mass of compound (Y) is ____ $gmol_{-1}$. (Nearest integer)

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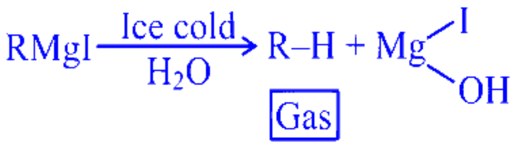
Answer: 30

Solution:

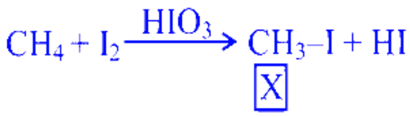


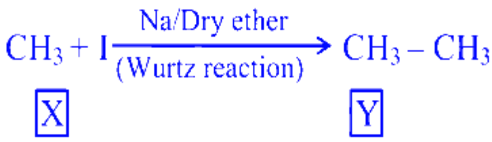
Given that the gas produced by treating RMgI with ice-cold water occupies a volume of $1.4\,\mathrm{dm}^3$ per gram at STP.

At STP, one mole of any gas occupies $22.4\,\mathrm{dm}^3$. So, the molar mass of the gas can be calculated as:

Molar mass $= \frac{22.4}{1.4} \approx 16$

This shows that the gas produced is methane, CH_4.





The methane formed reacts with iodine in the presence of HIO_3 to form methyl iodide (CH_3I), which is compound (X).

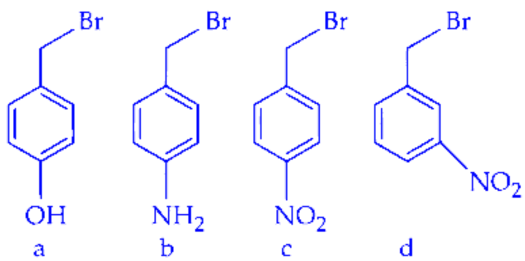
When CH_3I is treated with sodium metal in dry ether, a coupling reaction (Wurtz reaction) occurs, forming ethane (C_2H_6), which is compound (Y).

The molar mass of ethane is:

Molar mass of $C_2H_6 = (2 \times 12) + (6 \times 1) = 30\,\mathrm{g\,mol}^{-1}$

Question3

The correct order of reactivity of the following benzyl halides towards reaction with KCN is :



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Options:

A.

a > b > c > d

B.

b > a > d > c

C.

b > a > c > d

D.

a > b > d > c

Answer: B

Solution:

Type of reaction: This reaction happens by the S_N1 mechanism.

What affects the rate: The speed of an S_N1 reaction depends on how stable the carbocation (positively charged ion) formed in the process is.

Question4

The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is :

$$CH_3Br + Nu^- \longrightarrow CH_3Nu + Br^-$$

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Options:

A.

$ClO_4^- > CH_3COO^- > OH^- > PhO^-$

B.

$OH^- > PhO^- > CH_3COO^- > ClO_4^-$

C.

$CH_3COO^- > PhO^- > OH^- > ClO_4^-$

D.

$PhO^- > OH^- > CH_3COO^- > ClO_4^-$

Answer: B

Solution:

For CH_3Br (a methyl halide), the reaction proceeds by S_N2, so the **rate increases with nucleophilicity** of Nu^-.

For oxygen nucleophiles in the same row, nucleophilicity roughly follows **basicity**, and any **resonance stabilization** of the anion **reduces** nucleophilicity:

- OH^-: strong base, charge localized $\Rightarrow$ **best nucleophile**
- PhO^-: resonance-stabilized $\Rightarrow$ less nucleophilic than OH^-
- CH_3COO^-: even more resonance delocalization over two oxygens $\Rightarrow$ weaker nucleophile
- ClO_4^-: highly resonance-stabilized, essentially **non-nucleophilic**

$$OH^- \; > \; PhO^- \; > \; CH_3COO^- \; \gg \; ClO_4^-$$

Correct option: B.

Question5

The correct order of reactivity of $\mathrm{CH_3Br}$ in methanol with the following nucleophiles is

$\mathrm{F^-}, \mathrm{I^-}, \mathrm{C_2H_5O^-}$ **and** $\mathrm{C_6H_5O^-}$

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Options:

A.

$\mathrm{I^-} > \mathrm{C_2H_5O^-} > \mathrm{C_6H_5O^-} > \mathrm{F^-}$

B.

$\mathrm{I^-} > \mathrm{F^-} > \mathrm{C_6H_5O^-} > \mathrm{C_2H_5O^-}$

C.

$\mathrm{I^-} > \mathrm{C_6H_5O^-} > \mathrm{F^-} > \mathrm{C_2H_5O^-}$

D.

$\mathrm{I^-} > \mathrm{C_2H_5O^-} > \mathrm{F^-} > \mathrm{C_6H_5O^-}$

Answer: A

Solution:

In methanol, $\mathrm{CH_3Br}$ reacts by S_N2 mechanism, so the rate depends mainly on the nucleophilicity of the nucleophile.

In a protic solvent like methanol, smaller anions like $\mathrm{F^\ominus}$ are strongly solvated (surrounded by solvent molecules).
Because of this strong solvation, $\mathrm{F^\ominus}$ becomes less available to attack the carbon, so it is the weakest nucleophile.

Larger anions like $\mathrm{I^\ominus}$ are less solvated in methanol, so they can attack more easily. Hence, $\mathrm{I^\ominus}$ is the strongest nucleophile.

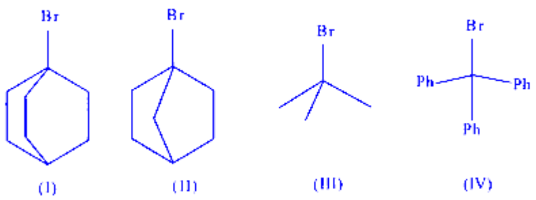
Among $\mathrm{C_2H_5O^\ominus}$ and $\mathrm{PhO^\ominus}$, phenoxide $(\mathrm{PhO^\ominus})$ is less nucleophilic.
This is because its negative charge is delocalised (resonance) into the benzene ring, reducing its tendency to attack.
Ethoxide $(\mathrm{C_2H_5O^\ominus})$ has a more localised negative charge, so it is a better nucleophile.

Order of nucleophilicity :
$\mathrm{I^\ominus} > \mathrm{C_2H_5O^\ominus} > \mathrm{PhO^\ominus} > \mathrm{F^\ominus}$

Question6

The correct order of the rate of reaction of the following reactants with nucleophile by $\mathrm{S_N1}$ mechanism is :

(Given : Structures I and II are rigid)



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Options:

A.

I < II < III < IV

B.

II < I < III < IV

C.

III < I < II < IV

D.

IV < III < II < I

Answer: B

Solution:

Rate of $S_N1 \propto$ Stability of $C^{\oplus}$ formed

(I) and (II) are unstable due to Bredt's rule, (I) has more + I effect.

$(\mathrm{II})<(\mathrm{I})<(\mathrm{III})<(\mathrm{IV})$

Question7

As compared with chlorocyclohexane, which of the following statements correctly apply to chlorobenzene?

A. The magnitude of negative charge is more on chlorine atom.

B. The $\mathrm{C}-\mathrm{Cl}$ bond has partial double bond character.

C. $\mathrm{C}-\mathrm{Cl}$ bond is less polar.

D. $\mathrm{C}-\mathrm{Cl}$ bond is longer due to repulsion between delocalised electrons of the aromatic ring and lone pairs of electrons of chlorine.

E. The $\mathrm{C}-\mathrm{Cl}$ bond is formed using sp^2 hybridised orbital of carbon.

Choose the correct answer from the options given below :

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Options:

A.

B, C and E Only

B.

A, C and E Only

C.

B, C and D Only

D.

A, D and E Only

Answer: A

Solution:

In chlorocyclohexane, chlorine shows only the $-\mathrm{I}$ (electron-withdrawing inductive) effect.

So, the $C-Cl$ bond remains a normal single bond and is more polar (more electron density is pulled towards chlorine).

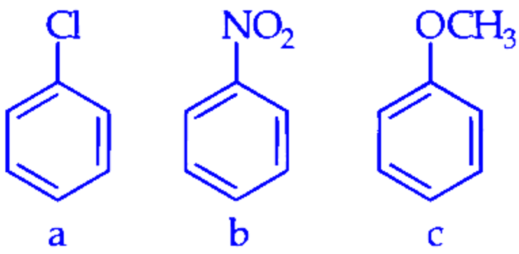
In chlorobenzene, chlorine shows two effects: $-I$ and $+M$ (lone pair donation by resonance into the benzene ring).

Because $-I > +M$, the bond is still polar, but the $+M$ effect reduces the polarity compared to chlorocyclohexane.

The $+M$ effect also causes resonance between chlorine lone pairs and the aromatic ring, which gives the $C-Cl$ bond partial double bond character.

Question8

Consider the following compounds



Arrange these compounds in the increasing order of reactivity with nitrating mixture.

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Options:

A.

$c < b < a$

B.

$b < c < a$

C.

$c < a < b$

D.

$b < a < c$

Answer: D

Solution:

In $Ph-OMe$, the $-OMe$ group is an electron-donating group and shows $+M$ (plus mesomeric) effect. It donates electron density to the benzene ring, increases the electron density on the ring, and therefore **activates** the ring towards electrophilic substitution such as nitration.

In $Ph-NO_2$, the $-NO_2$ group is a strong electron-withdrawing group and shows $-M$ (minus mesomeric) effect. It pulls electron density away from the benzene ring and therefore **strongly deactivates** the ring towards nitration.

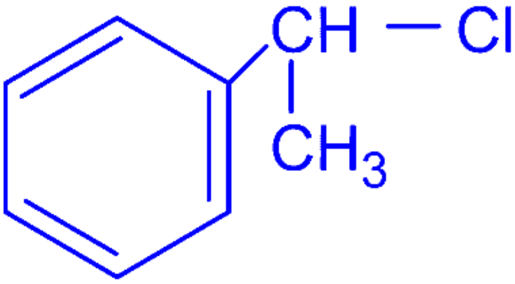
In $Ph-Cl$, the $-Cl$ group is overall an electron-withdrawing group (due to its $-I$ or inductive effect). Because of this electron-withdrawing nature, it **decreases** the reactivity of the benzene ring towards nitration, but this effect is weaker than that of the $-NO_2$ group.

Therefore, the increasing order of reactivity towards nitration is: $Ph-NO_2 < Ph-Cl < Ph-OMe$.

Question9

Match the List-I with List-II

List-I Chloro derivative	List-II Example
A. Vinyl Chloride	I. $CH_2 = CH - CH_2Cl$

B. Benzyl Chloride	II. CH_3 - $CH(Cl)CH_3$
C. Alkyl Chloride	III. $CH_2 = CHCl$
D. Allyl Chloride	IV. 

Choose the correct answer from the options given below :

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Options:

A.

A-III, B-IV, C-I, D-II

B.

A-IV, B-I, C-III, D-II

C.

A-III, B-IV, C-II, D-I

D.

A-I, B-II, C-IV, D-III

Answer: C

Solution:

Vinyl, allyl, benzyl and alkyl chlorides are identified from the position of Cl:

Step 1: Identify each example in List-II

1. **III.** $CH_2 = CHCl$

   Here Cl is directly attached to a double-bond carbon $\Rightarrow$ **vinyl chloride**.

2. **I.** $CH_2 = CH - CH_2Cl$

   Here Cl is on the carbon next to $C = C$ (allylic position) $\Rightarrow$ **allyl chloride**.

3. **II.** $CH_3 - CH(Cl)CH_3$

   This is a saturated alkyl group with Cl on an sp^3 carbon $\Rightarrow$ **alkyl chloride** (2-chloropropane).

4. **IV.** (structure shown is $C_6H_5 - CH_2Cl$)

   Cl is on the carbon next to benzene ring (benzylic position) $\Rightarrow$ **benzyl chloride**.

Step 2: Match List-I with List-II

- **A. Vinyl chloride $\rightarrow$ III**
- **B. Benzyl chloride $\rightarrow$ IV**
- **C. Alkyl chloride $\rightarrow$ II**
- **D. Allyl chloride $\rightarrow$ I**

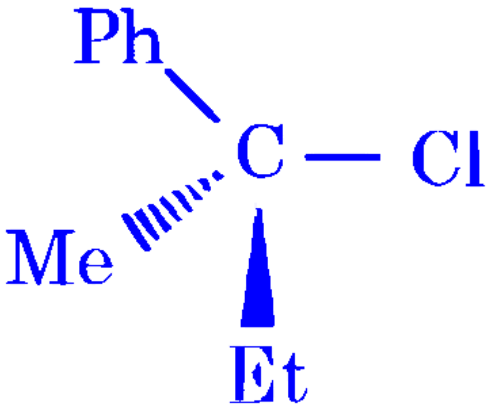
So the correct option is:

Option C: A-III, B-IV, C-II, D-I

Question10

Given below are two statements :

Statement I : ' $\mathrm{C-Cl}$ **' bond is stronger in** $\mathrm{CH_2=CH-Cl}$ **than** $\mathrm{CH_3-CH_2-Cl}$

Statement II : The given optically active molecule,  **on hydrolysis gives a solution that can rotate the plane polarized light.**

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Statement I is false but Statement II is true

B.

Statement I is true but Statement II is false

C.

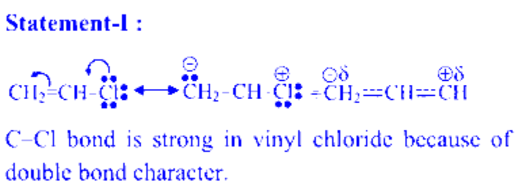
Both Statement I and Statement II are true

D.

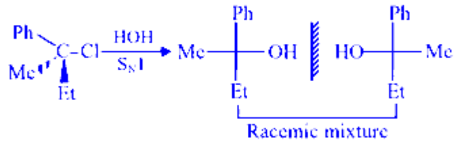
Both Statement I and Statement II are false

Answer: B

Solution:



Statement-II :



Racemic mixture is optically inactive, which can not rotate PPL.

Question11

Consider the following reactions giving major product. Identify the correct reaction.

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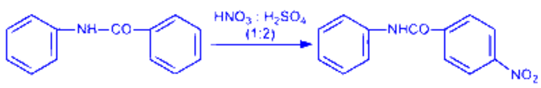
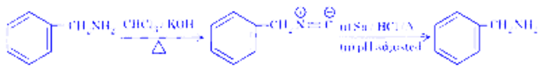
Options:

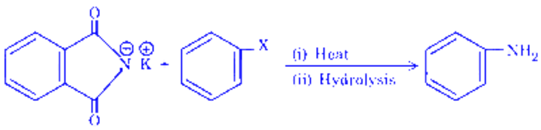
A.



B.



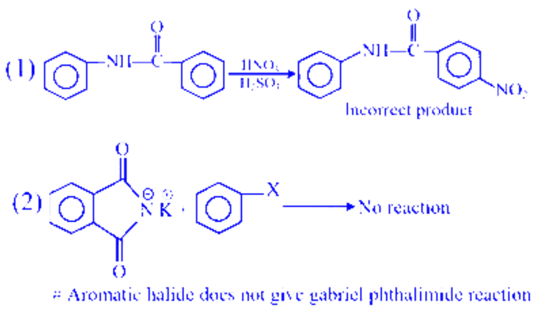
C.

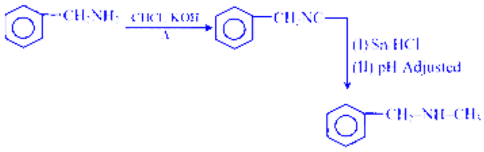


D.

Answer: D

Solution:



(3) 

(4) Hoffmann bromamide degradation

$$CH_3-CH_2-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2 \xrightarrow[4KOH]{Br_2} CH_3-CH_2-NH_2 + 2KBr + K_2CO_3 + 2H_2O$$

Question12

$\mathrm{Ph-CH=CH_2} \xrightarrow[\mathrm{HBr}]{(\mathrm{PhCOO})_2}$ **Product**

Consider the above reaction

A. The reaction proceeds through a more stable radical intermediate.

B. The role of peroxide is to generate $H^\bullet$ (Hydrogen radical).

C. During this reaction, benzene is formed as a byproduct.

D. 1-Bromo-2-phenylethane is formed as the minor product.

E. The same reaction in absence of peroxide proceeds via carbocation intermediate.

Identify the correct statements. Choose the correct answer from the options given below :

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Options:

A.

A & E Only

B.

A, B & D Only

C.

A, C & E Only

D.

C, D & E Only

Answer: C

Solution:

In presence of peroxide, addition of HBr to an alkene follows **free radical mechanism** (peroxide effect).

For styrene $\mathrm{Ph-CH=CH_2}$:

Major product (with peroxide)

Br• adds in such a way that the **more stable radical** is formed.

If Br• adds to terminal carbon (CH_2), the radical comes on benzylic carbon:

$\mathrm{Ph{-}\dot{C}H{-}CH_2Br}$

This is **benzylic radical** (resonance-stabilised), hence preferred. Then it abstracts H from HBr:

$\mathrm{Ph{-}CH_2{-}CH_2Br}$

So **1-bromo-2-phenylethane** is the **major** product.

Checking each statement

A. True

The reaction proceeds via the **more stable radical intermediate** (benzylic radical).

B. False

Peroxide does **not** generate $H^{\bullet}$.

It generates radicals that ultimately produce $Br^{\bullet}$ (the chain-carrying radical), e.g. by abstraction of H from HBr.

C. True

Benzoyl peroxide can form phenyl radicals ($Ph^{\bullet}$), which can abstract H from HBr:

$\mathrm{Ph^{\bullet} + HBr \rightarrow PhH + Br^{\bullet}}$

So **benzene (PhH)** can be formed as a byproduct.

D. False

1-bromo-2-phenylethane ($\mathrm{Ph{-}CH_2{-}CH_2Br}$) is the **major**, not minor, product in presence of peroxide.

E. True

In absence of peroxide, HBr adds by **electrophilic addition** via a **carbocation intermediate** (here benzylic carbocation), giving Markovnikov product.

Correct option: Option C (A, C & E Only)

Question13

Given below are two statements :

Statement (I) : Benzyl chloride reacts faster in S_N1 mechanism than ethyl chloride.

Statement (II) : Ethyl carbocation intermediate is less stabilized by hyperconjugation than benzyl carbocation by resonance.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is true but **Statement II** is false

D.

Statement I is false but **Statement II** is true

Answer: A

Solution:

In an S_N1 reaction, the rate depends on the stability of the carbocation formed after the leaving group departs.

For benzyl chloride:

$C_6H_5CH_2Cl \rightarrow C_6H_5CH_2^+ + Cl^-$

The benzyl carbocation formed is highly stable because the positive charge is delocalised over the benzene ring by resonance.

For ethyl chloride:

$CH_3CH_2Cl \rightarrow CH_3CH_2^+ + Cl^-$

The ethyl carbocation is much less stable. It gets only slight stabilization by hyperconjugation, which is weaker than resonance stabilization.

So:

- **Statement (I)** is true: benzyl chloride reacts faster than ethyl chloride in the S_N1 mechanism.
- **Statement (II)** is also true: ethyl carbocation is less stabilized by hyperconjugation than benzyl carbocation by resonance.

Also, Statement (II) correctly explains Statement (I).

Therefore, the correct answer is:

$\boxed{\text{Option A}}$

Question14

Correct statements regarding alkyl halides (R–X) among the following are :

A. Alcohol being less polar solvent as compared to water, alcoholic KOH favours elimination reaction with R–X.

B. Order of reactivity towards S_N1 mechanism is $C_6H_5–CH_2–Cl > C_6H_5–CHCl–C_6H_5$.

C. Non substituted aryl halides exhibit properties similar to alkyl halides.

D. Vinyl chloride is an example of haloalkene and allyl chloride is an example of haloalkyne.

E. R–Cl can be prepared by reacting R–OH with $SOCl_2$ but Ar–Cl cannot be prepared by reacting Ar–OH with $SOCl_2$.

Choose the correct answer from the options given below :

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Options:

A.

A, B and C Only

B.

B and D Only

C.

A and E Only

D.

D and E Only

Answer: C

Solution:

Check each statement carefully.

For $R-X$, alcoholic KOH generally gives elimination, while aqueous KOH gives substitution.

This happens because alcohol is less polar than water, so the hydroxide ion is less solvated and behaves more like a base, favouring β-elimination.

So, A is correct.

Now B:

In S_N1 reaction, reactivity depends on stability of the carbocation formed.

For benzyl chloride:

$C_6H_5-CH_2-Cl \rightarrow C_6H_5-CH_2^+$

This is a benzyl carbocation, stabilized by resonance.

For benzhydryl chloride:

$C_6H_5-CHCl-C_6H_5 \rightarrow C_6H_5-CH^+-C_6H_5$

This carbocation is stabilized by resonance with two phenyl rings, so it is more stable than benzyl carbocation.

Hence reactivity order should be

$C_6H_5-CHCl-C_6H_5 > C_6H_5-CH_2-Cl$

So statement B is false.

Now C:

Aryl halides do not show properties similar to alkyl halides.

They are much less reactive in nucleophilic substitution because the $C-X$ bond in aryl halides has partial double bond character due to resonance.

So C is false.

Now D:

Vinyl chloride, $CH_2=CHCl$, is a haloalkene. This part is correct.

Allyl chloride, $CH_2=CH-CH_2Cl$, is also a haloalkene, not a haloalkyne.

So D is false.

Now E:

Alkyl chlorides can be prepared from alcohols using thionyl chloride:

$R-OH+SOCl_2 \rightarrow R-Cl+SO_2+HCl$

But phenols do not give aryl chlorides with $SOCl_2$.

In phenol, the $C-O$ bond has partial double bond character due to resonance, so replacement of OH by Cl does not happen this way.

So E is correct.

The correct statements are A and E only.

$\boxed{\text{Option C}}$

Question15

Match the LIST-I with LIST-II

List-I Name of reaction		List-II Reagent or catalyst used	
A.	Finkelstein reaction	I.	SbF_3
B.	Swarts reaction	II.	Na, dry ether
C.	Sandmeyer's reaction	III.	NaI
D.	Fittig reaction	IV.	Cu_2Cl_2

Choose the correct answer from the options given below :

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Options:

A.

A-I, B-IV, C-III, D-III

B.

A-III, B-I, C-IV, D-II

C.

A-IV, B-II, C-I, D-III

D.

A-I, B-III, C-II D-IV

Answer: B

Solution:

A. Finkelstein reaction:

This is a halogen exchange reaction used specifically for the preparation of alkyl iodides. In this reaction, alkyl chlorides or bromides are treated with sodium iodide (NaI) in dry acetone.

Therefore, A matches with III.

B. Swarts reaction:

This reaction is the preferred method for the synthesis of alkyl fluorides. It involves heating an alkyl chloride or bromide in the presence of a metallic fluoride. Some of the common fluorinating agents used are AgF, Hg_2F_2, CoF_2, or antimony trifluoride (SbF_3).

Therefore, B matches with I.

C. Sandmeyer's reaction:

This reaction is used to prepare haloarenes (like chlorobenzene or bromobenzene) from benzene diazonium chloride. The diazonium salt is treated with a cuprous halide, such as cuprous chloride (Cu_2Cl_2 or $CuCl$), to replace the diazonium group with a halogen.

Therefore, C matches with IV.

D. Fittig reaction:

This reaction involves aryl halides. When an aryl halide is treated with sodium (Na) in the presence of dry ether, two aryl groups join together to form a diaryl compound (like biphenyl).

Therefore, D matches with II.

Summary of Matches:

- A (Finkelstein reaction) $\rightarrow$ III (NaI)
- B (Swarts reaction) $\rightarrow$ I (SbF_3)
- C (Sandmeyer's reaction) $\rightarrow$ IV (Cu_2Cl_2)
- D (Fittig reaction) $\rightarrow$ II (Na, dry ether)

Comparing our findings with the given options, the correct combination is A-III, B-I, C-IV, D-II.

Option B is the correct answer.

Question16

Given below are two statements :

Statement I : 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

Statement II : Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

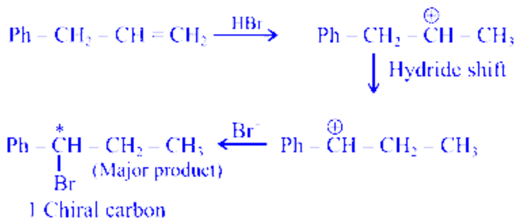
Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: C

Solution:



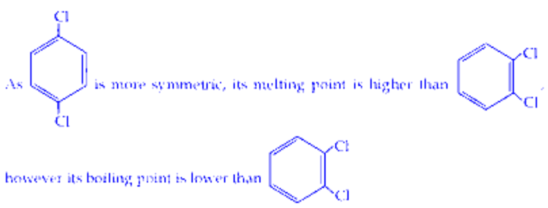
Aryl cyanide is not formed by Gattermann reaction.

Question17

Given below are two statements :

Statement I : Due to increase in van der Waals forces, the order of boiling points is $\mathrm{CH_3CH_2CH_2I > CH_3CH_2I > CH_3I}$.

Statement II :



In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

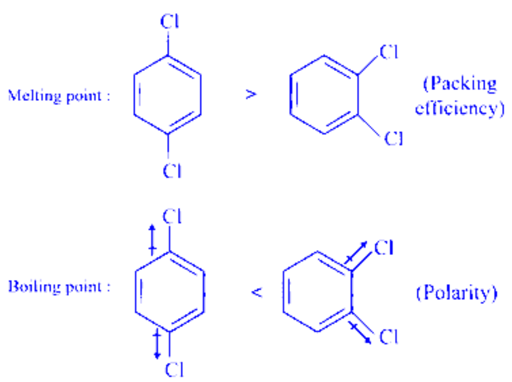
Answer: A

Solution:

6. As the hydrocarbon chain increases, effective van der waals force also increases. This results in increase in boiling point also.

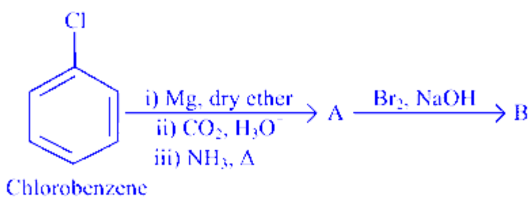
$CH_3 - CH_2 - CH_2 - I > CH_3 - CH_2 - I > CH_3 - I.$

Para substituted chlorobenzene has greater packing than ortho substituted. Better packing leads to higher melting point. Ortho-substituted chlorobenzene has greater dipole - dipole interaction that para one which gives higher boiling point.



Question18

Consider the following sequence of reactions:



11.25 mg of chlorobenzene will produce ___________$\times 10^{-1}$mg of product B.

(Consider the reactions result in complete conversion.)

[Given molar mass of C, H, O, N and Cl as $12, 1, 16, 14$ and 35.5 g mol^{-1} respectively]

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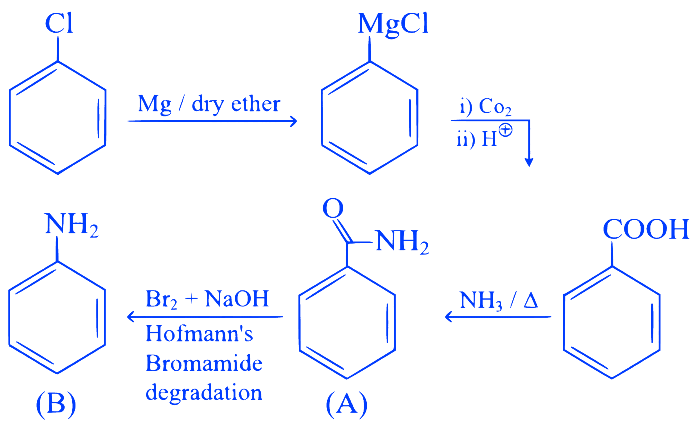
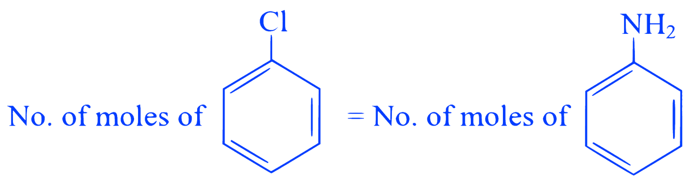
Answer: 93

Solution:





$$\frac{11.25 \times 10^{-3}}{112.5} = \frac{x \times 10^{-1} \times 10^{-3}}{93}$$
$x \times 10^{-1} = 93 \times 0.1$
$x = 93\mathrm{mg}$

Question19

Consider the above sequence of reactions. 151 g of 2-bromopentane is made to react. Yield of major product P is 80% **whereas Q is** 100%**.**

Mass of product Q obtained is________ g.

(Given molar mass in $\mathrm{gmol}^{-1} \mathrm{H} : 1, \mathrm{C} : 12, \mathrm{O} : 16, \mathrm{Br} : 80$ **)**

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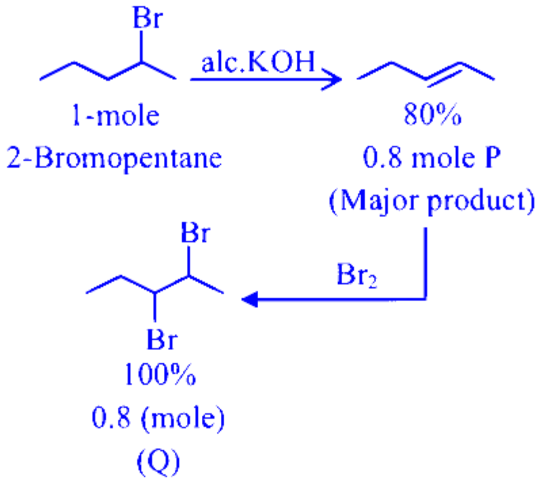
Answer: 184

Solution:



Molecular mass of $Q = 230 \text{ g mol}^{-1}$

Mass of $Q = 0.8 \times 230$

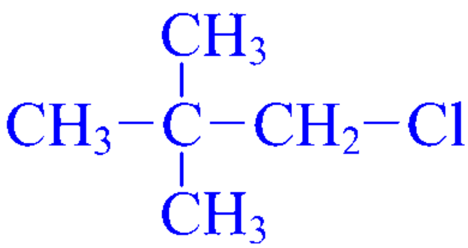
$= 184 \text{ g}$

Question20

Given below are two statements :

Statement I : $CH_3 - O - CH_2 - Cl$ **will undergo** S_N1 **reaction though it is a primary halide.**

Statement II :



will not undergo S_N2 **reaction very easily though it is a primary halide.**

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A. Both Statement I and Statement II are incorrect

B. Both Statement I and Statement II are correct

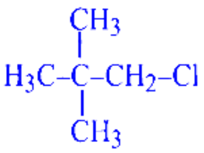
C. Statement I is correct but Statement II is incorrect

D. Statement I is incorrect but Statement II is correct

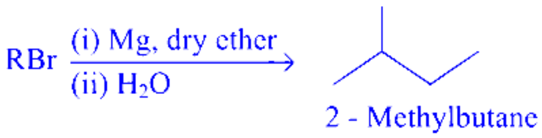
Answer: B

Solution:

$CH_3 - O - CH_2 - Cl$ will undergo S_N1 mechanism because $CH_3 - O - \overset{+}{C}H_2$ is highly stable.



Question21



The maximum number of RBr producing 2-methylbutane by above sequence of reactions is___________ . (Consider the structural isomers only)

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Options:

A. 5

B. 3

C. 4

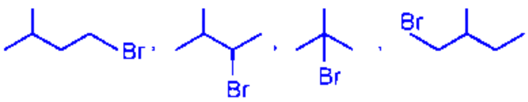
D. 1

Answer: C

Solution:

$RBr \xrightarrow[\text{dry Ether}]{Mg} RMgBr \xrightarrow{H_2O} R - H$

Hence, RBr Can be



Total 4 Structural isomers.

Question22

Propane molecule on chlorination under photochemical condition gives two di-chloro products, " x " and " y ". Amongst " x " and " y ", " x " is an optically active molecule. How many tri-chloro products (consider only structural isomers) will be obtained from " x " when it is further treated with chlorine under the photochemical condition?

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Options:

A. 4

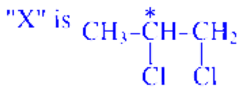
B. 2

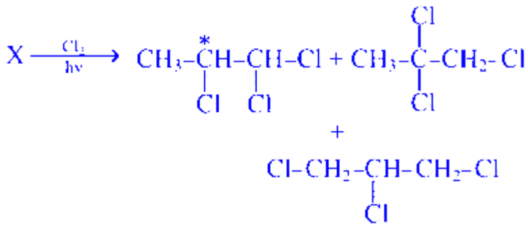
C. 3

D. 5

Answer: C

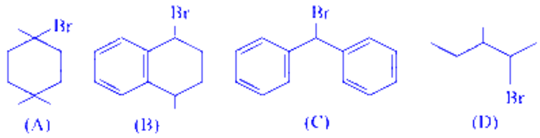
Solution:





Question23

The ascending order of relative rate of solvolysis of following compounds is :



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Options:

A. (C) < (D) < (B) < (A)

B. (D) < (A) < (B) < (C)

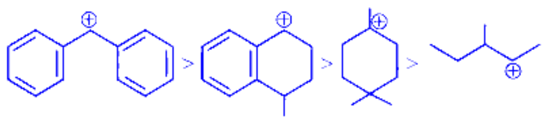
C. (D) < (B) < (A) < (C)

D. (C) < (B) < (A) < (D)

Answer: B

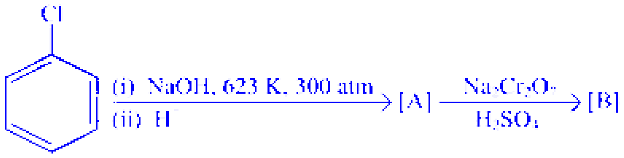
Solution:

Solvolysis or $S_N1 \propto$ stability of carboccation Stability order



Question24

Identify the products [A] and [B], respectively in the following reaction:



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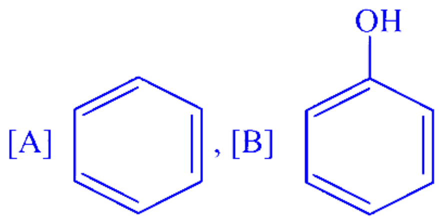
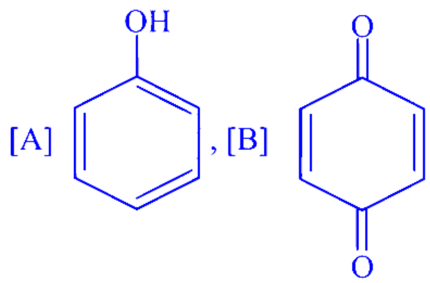
Options:

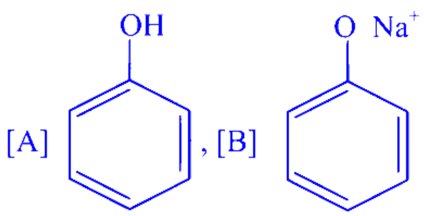
A.



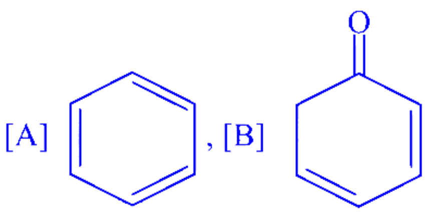
B.



C.



D.



Answer: B

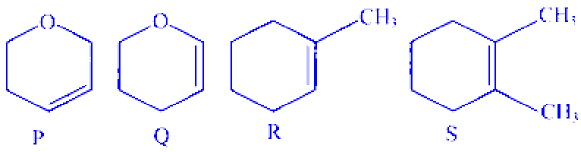
Solution:

A is phenol and B is para benzoquinone.

Question25

Following are the four molecules "P", "Q", "R" and "S".

Which one among the four molecules will react with $\mathrm{H-Br_{(aq)}}$ at the fastest rate?



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Options:

A. Q

B. S

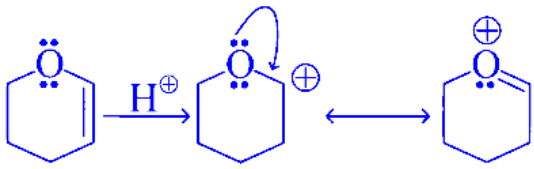
C. P

D. R

Answer: A

Solution:

Addition of $\mathrm{H-Br_{(aq)}}$ to alkene follows electrophilic addition mechanism. In the rate determining step a carbocation intermediate is formed. Among P, Q, R & S compound Q will form most stable carbocation intermediate since it is resonance stabilized.



Question26

Given below are two statements:

Statement I: The conversion proceeds well in the less polar medium.

$$\mathrm{CH_3-CH_2-CH_2-CH_2-Cl \xrightarrow{HO^-} CH_3-CH_2-CH_2-CH_2-OH + Cl^{(-)}}$$

Statement II: The conversion proceeds well in the more polar medium.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A. Statement I is false but Statement II is true

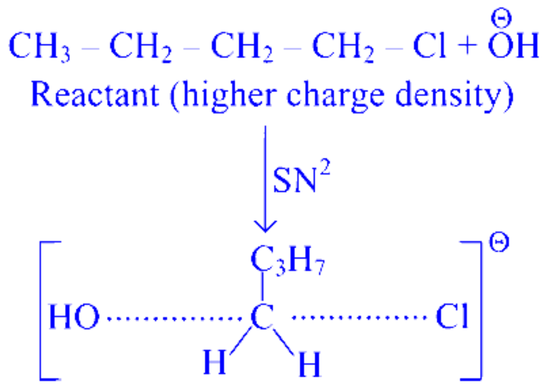
B. Statement I is true but Statement II is false

C. Both Statement I and Statement II are false

D. Both Statement I and Statement II are true

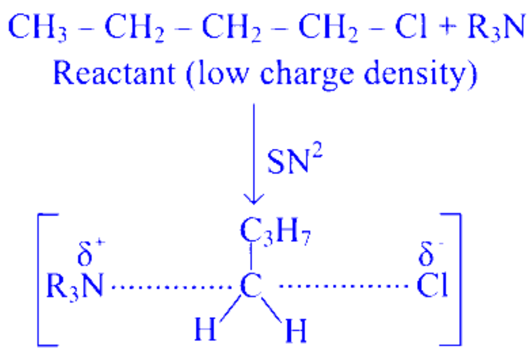
Answer: D

Solution:



Transition state (less charge density)

⇒ This reaction will proceed faster in less polar medium which will not increase the activation energy value.

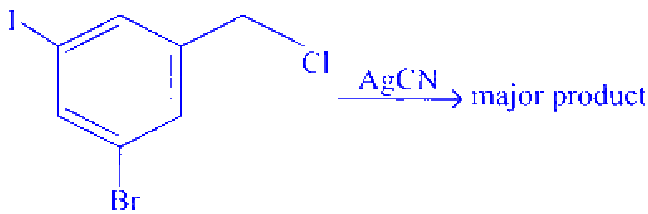


Transition state (Higher charge density)

⇒ This reaction will proceed faster in more polar medium which will decrease the activation energy value.

Question27

The structure of the major product formed in the following reaction is :



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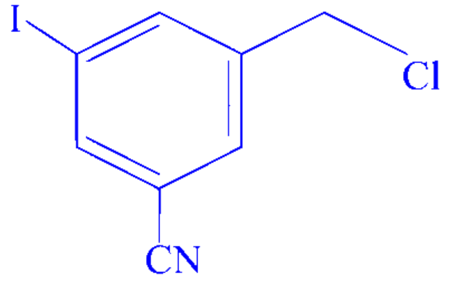
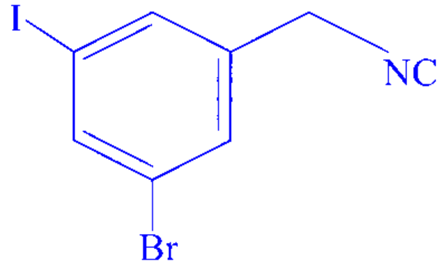
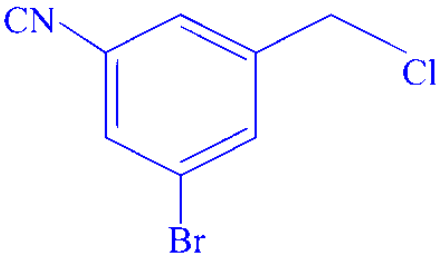
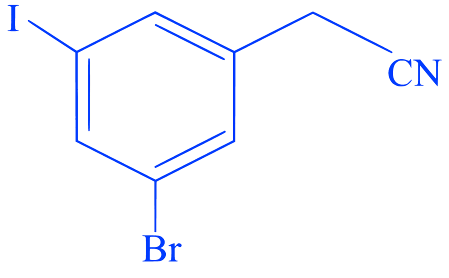
Options:

A. 

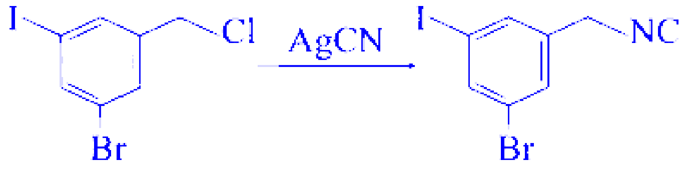
B. 

C. 

D. 

Answer: B

Solution:



Question28

The products A and B in the following reactions, respectively are

$$A \overset{Ag-NO_2}{\longleftrightarrow} CH_3 - CH_2 - CH_2 - Br \xrightarrow{AgCN} B$$

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Options:

A. $CH_3 - CH_2 - CH_2 - ONO, CH_3 - CH_2 - CH_2 - CN$

B. $CH_3 - CH_2 - CH_2 - NO_2, CH_3 - CH_2 - CH_2 - CN$

C. $CH_3 - CH_2 - CH_2 - ONO, CH_3 - CH_2 - CH_2 - NC$

D. $CH_3 - CH_2 - CH_2 - NO_2, CH_3 - CH_2 - CH_2 - NC$

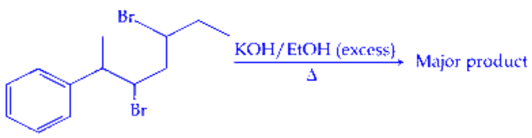
Answer: D

Solution:

$\underset{(A)}{CH_3 - CH_2 - CH_2 - NO_2} \overset{Ag-NO_2}{\longrightarrow} CH_3 - CH_2 - CH_2 - Br \overset{AgCN}{\longrightarrow} \underset{(B)}{CH_3 - CH_2 - CH_2 - NC}$

Question29

The major product of the following reaction is:



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Options:

A.

6-Phenylhepta-3,5-diene

B.

6-Phenylhepta-2,4-diene

C.

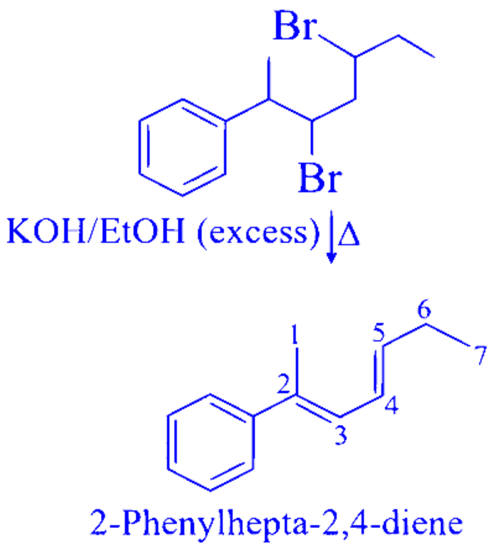
2-Phenylhepta-2,4-diene

D.

2-Phenylhepta-2,5-diene

Answer: C

Solution:



Question30

Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?

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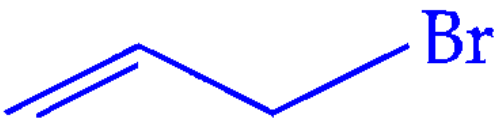
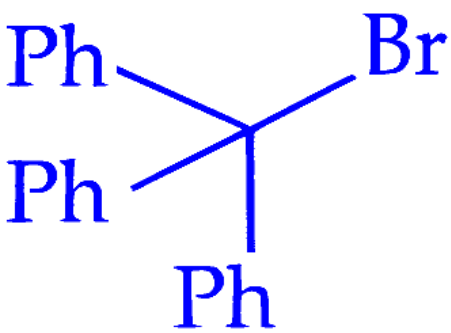
Options:

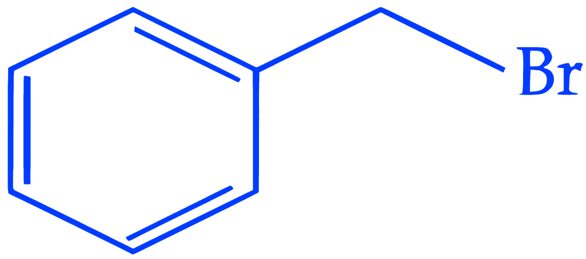
A.



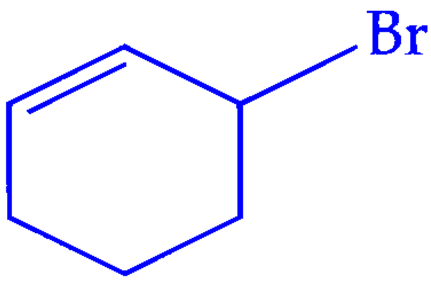
B.



C.



D.



Answer: B

Solution:

A nucleophilic substitution reaction is a chemical reaction in which an electron rich molecule (nucleophile) replaces a functional group in an electron deficient molecule (electrophile). Carbocation formation occurs in SN^1 mechanism (unimolecular nucleophilic substitution).

Carbocation is formed in the first step when the bond between carbon and the leaving group breaks. The nucleophile attack on the carbocation occurs in the second step.

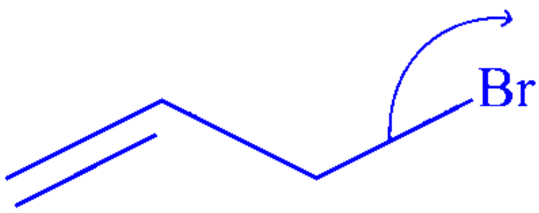
The general stability order of carbocation is

Tertiory > Secondary > Primary

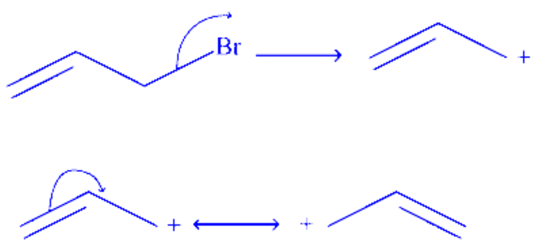
The stability order including allyl carbocation, benzyl carbocation, phenyl carbocation and triphenyl methyl $((Ph)_3C^+)$ carbocation is

Phenyl < Allyl < Benzyl < Triphenyl methyl

(1)

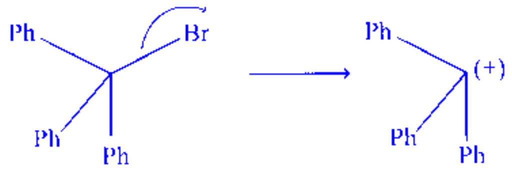


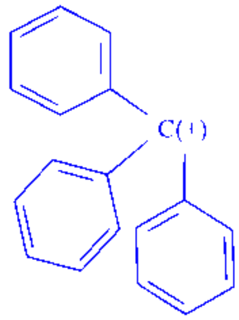
Carbocation formed from this halide is



Allylic carbocation stability is due to the resonance delocalization of the positive change across the adjacent double bond.

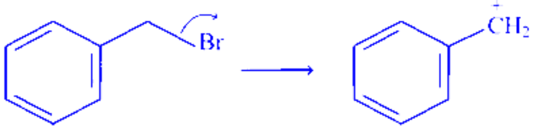
(2)





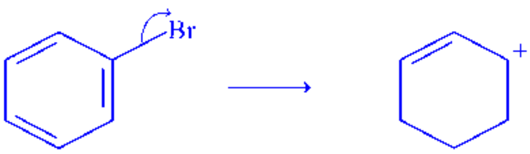
Stable due to the delocalization of positive change across all three phenyl rings.

(3)

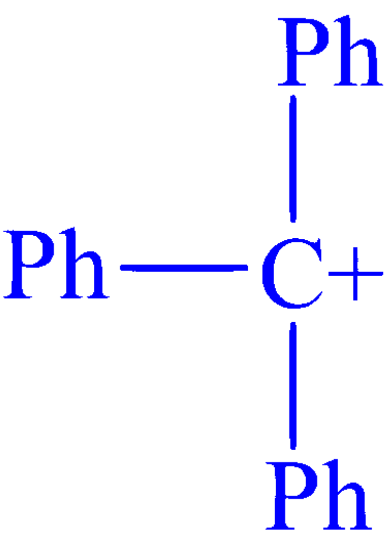


Benzyl carbocation stability is because of the positive charge delocalization across the aromatic ring. So, it is mole stable than allylic carbocation but less stable than Ph_3C+ carbocation.

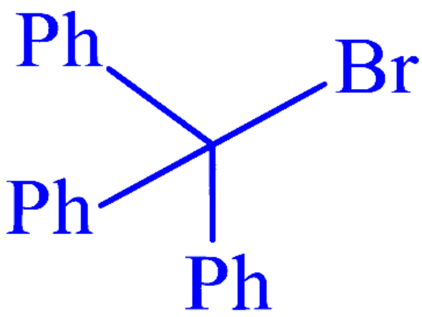
(4)



Delocalization is possible for the adjacent double bond. So, most stable carbocation is

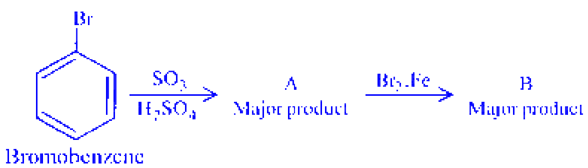


The answer is option (2)



Question31

In the following series of reactions identify the major products A&B respectively



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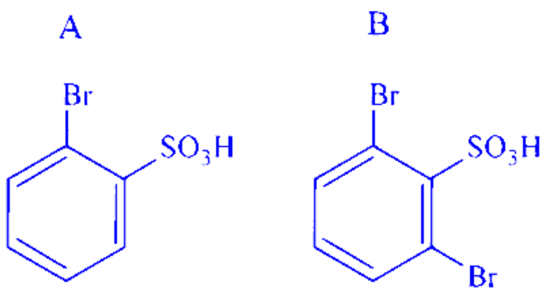
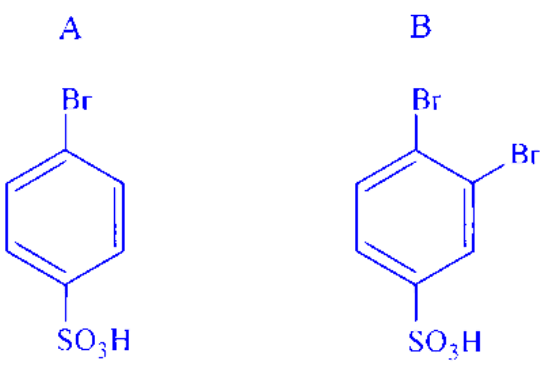
Options:

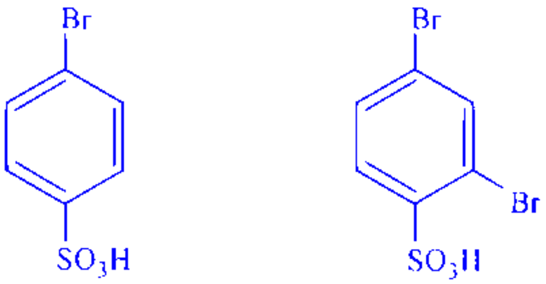
A.



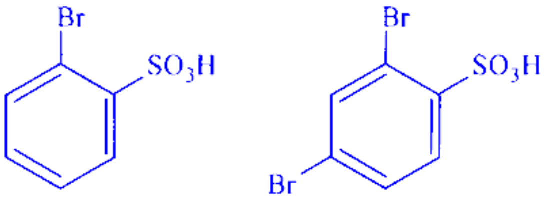
B.



C. A B

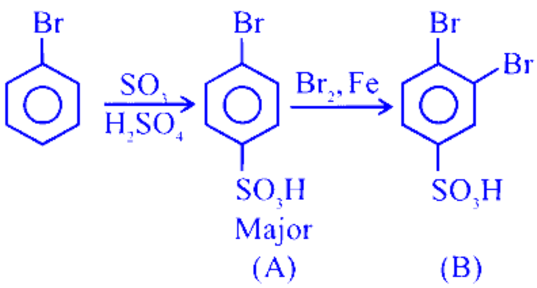


D. A B



Answer: B

Solution:



Both reactions are electrophilic substitution reaction, I^{st} is sulphonation and II^{nd} is halogenation :

Question32

Given below are two statements :

Statement (I) : Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.

Statement (II) : In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the β-carbon.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A. Both Statement I and Statement II are incorrect

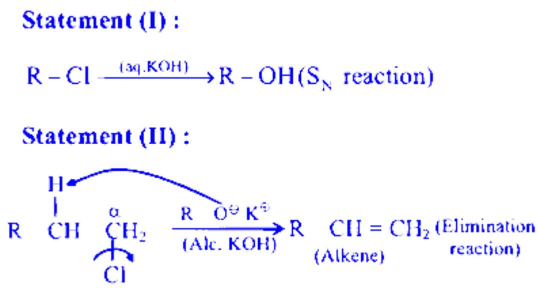
B. Both Statement I and Statement II are correct

C. Statement I is incorrect but Statement II is correct

D. Statement I is correct but Statement II is incorrect

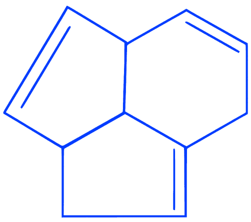
Answer: C

Solution:



Question33

Consider the following molecule(X).

The structure of X is 

(X)

The major product formed when the given molecule (*X*) is treated with HBr(1eq) is :

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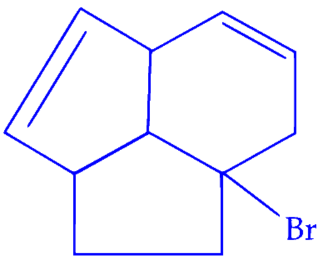
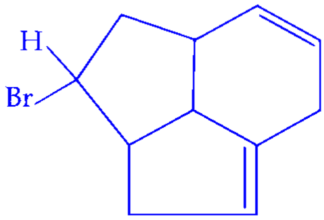
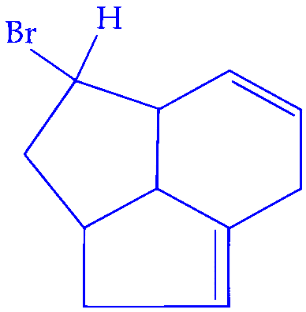
Options:

A. 

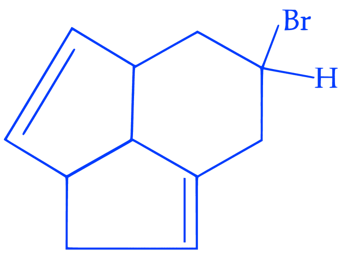
B.



C.



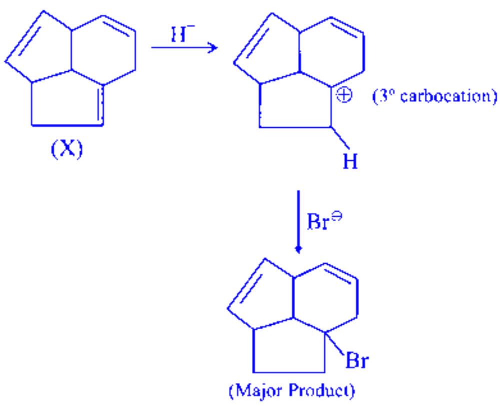
D.



Answer: A

Solution:

Among the given options, the major product decided by stability of carbocation formed inintermediate.



Question34

Which one of the following reactions will not lead to the desired ether formation in major proportion?

(iso- Bu $\Rightarrow$ isobutyl, sec $-$Bu $\Rightarrow$ sec-butyl, nPr $\Rightarrow$ n-propyl, ^{t}Bu $\Rightarrow$ tert-butyl, Et $\Rightarrow$ ethyl)

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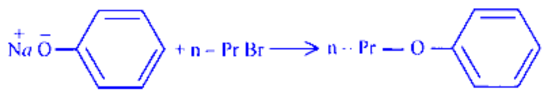
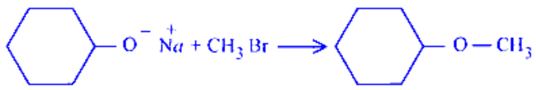
Options:

A. 

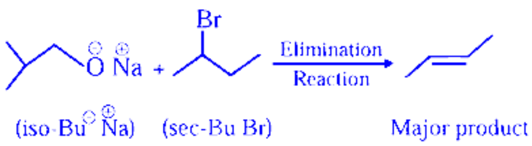
B. iso- $Bu\overline{O}\overset{+}{Na}$ +sec – BuBr $\longrightarrow$ sec- Bu – O– iso -Bu

C. 

D. ${}^tBu\overline{O}\overset{+}{Na}$ +EtBr $\longrightarrow$ ${}^tBu - O - Et$

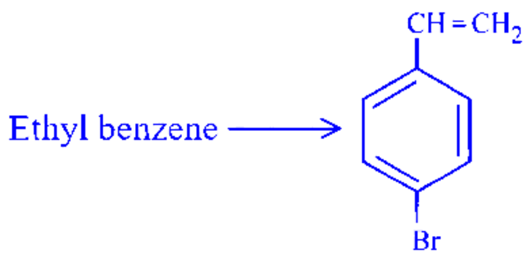
Answer: B

Solution:



Question35

Choose the correct set of reagents for the following conversion.



JEE Main 2025 (Online) 8th April Evening Shift

Options:

A. $Cl_2/$ anhy $\cdot AlCl_3; Br_2/Fe;$ alc. KOH

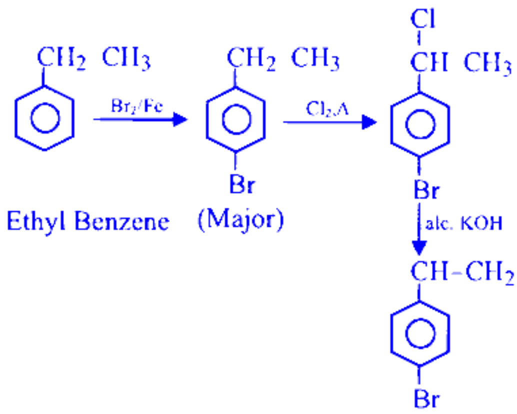
B. $Cl_2/Fe; Br_2/$ anhy. $AlCl_3;$ aq. KOH

C. $Br_2/Fe; Cl_2, \Delta;$ alc. KOH

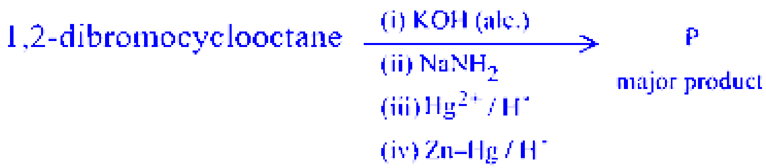
D. $Br_2/$ anhy. $AlCl_3; Cl_2, \Delta;$ aq. KOH

Answer: C

Solution:



Question36



'P' is

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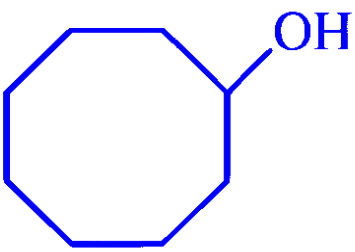
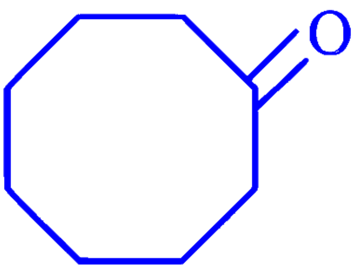
Options:

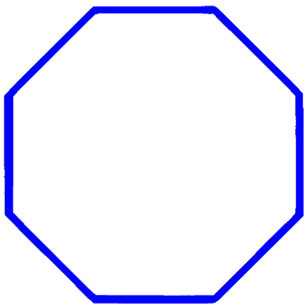
A.



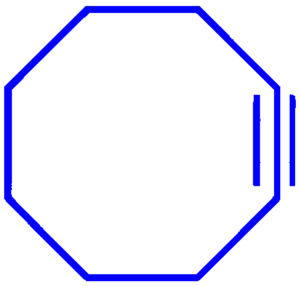
B.



C.

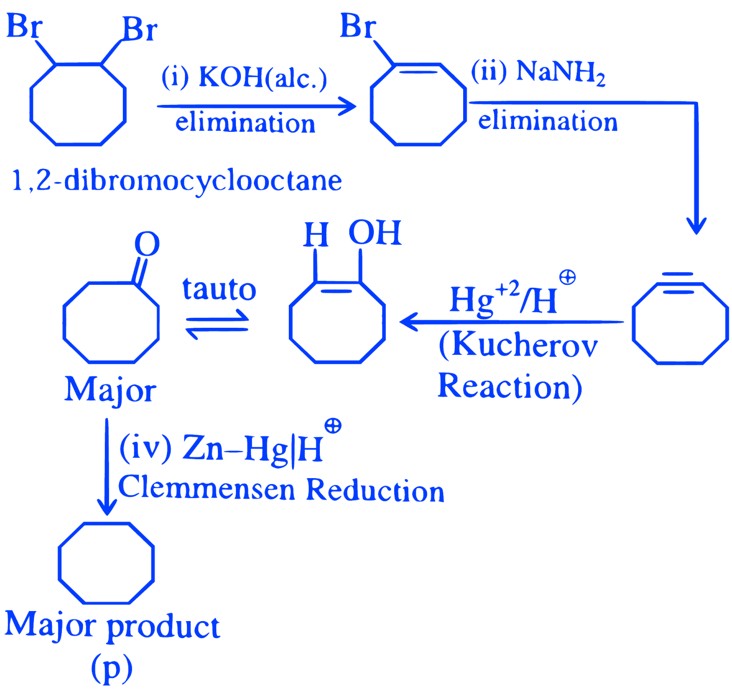


D.



Answer: C

Solution:



Question37

The correct statement regarding nucleophilic substitution reaction in a chiral alkyl halide is ; [27-Jan-2024 Shift 1]

Options:

A. Retention occurs in S_N1 reaction and inversion occurs in S_N2 reaction.

B. Racemisation occurs in S_N1 reaction and retention occurs in S_N2 reaction.

C. Racemisation occurs in both S_N1 and S_N2 reactions.

D. Racemisation occurs in S_N1 reaction and inversion occurs in S_N2 reaction.

Answer: D

Question38

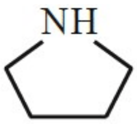
Identify B formed in the reaction.

$$\mathbf{Cl-(CH_2)_4-Cl \xrightarrow{\text{excess } NH_3} A \xrightarrow{NaOH} B + H_2O + NaCl}$$

[27-Jan-2024 Shift 2]

Options:

A.



B. $H_2N-(CH_2)_4-NH_2$

C. $Cl\overset{+}{N}H_3-(CH_2)_4-\overset{+}{N}H_3Cl^-$

D.

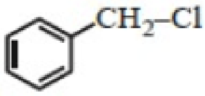
Answer: B

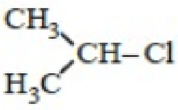
Question39

Which among the following halide/s will not show S_N1 reaction:

(A) $H_2C = CH - CH_2\,Cl$

(B) $CH_3 - CH = CH - Cl$

(C) 

(D) 

Choose the most appropriate answer from the options given below:
[27-Jan-2024 Shift 2]

Options:

A. (A), (B) and (D) only

B. (A) and (B) only

C. (B) and (C) only

D. (B) only

Answer: D

Question40

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :
Assertion A : Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.
Reason R : Phenols react with halogen acids violently. In the light of the above statements, choose the most appropriate from the options given below:
[29-Jan-2024 Shift 1]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

B. A is false but R is true

C. A is true but R is false

D. Both A and R are true and R is the correct explanation of A

Answer: C

Question41

Alkyl halide is converted into alkyl isocyanide by reaction with [29-Jan-2024 Shift 2]

Options:

A. NaCN

B. NH_4CN

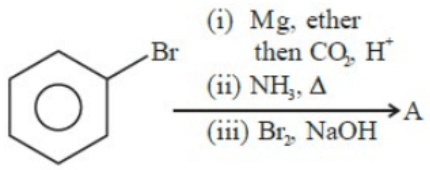
C. KCN

D. AgCN

Answer: D

Question42

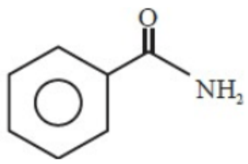
The final product A, formed in the following multistep reaction sequence is:



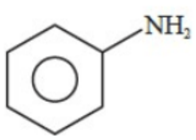
[30-Jan-2024 Shift 1]

Options:

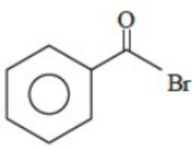
A.



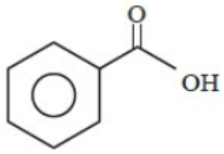
B.



C.



D.



Answer: B

Question43

Given below are two statements:
Statement - I: High concentration of strong nucleophilic reagent with secondary alkyl halides which do not have bulky substituents will follow S_N2 mechanism.
Statement - II: A secondary alkyl halide when treated with a large excess of ethanol follows S_N1 mechanism.
In the the light of the above statements, choose the most appropriate from the questions given below:
[30-Jan-2024 Shift 2]

Options:

A. Statement I is true but Statement II is false.

B. Statement I is false but Statement II is true.

C. Both statement I and Statement II are false.

D. Both statement I and Statement II are true.

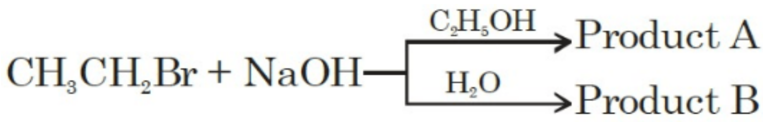
Answer: D

Question44

2-chlorobutane $+Cl_2 \rightarrow C_4H_8Cl_2$ (isomers)
Total number of optically active isomers shown by $C_4H_8Cl_2$, obtained in the above reaction is____
[30-Jan-2024 Shift 2]

Answer: 6

Question45

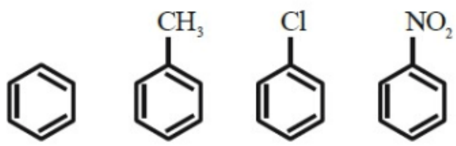


The total number of hydrogen atoms in product A and product B is_____
[31-Jan-2024 Shift 1]

Answer: 10

Question46

The correct order of reactivity in electrophilic substitution reaction of the following compounds is :



A B C D

[31-Jan-2024 Shift 2]

Options:

A. B > C > A > D

B. D > C > B > A

C. A > B > C > D

D. B > A > C > D

Answer: D

Question47

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Haloalkanes react with KCN to form alkyl cyanides as a main product while with AgCN form isocyanide as the main product.
Reason (R) : KCN and AgCN both are highly ionic compounds.
In the light of the above statement, choose the most appropriate answer from the options given below:
[1-Feb-2024 Shift 1]

Options:

A. (A) is correct but (R) is not correct

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

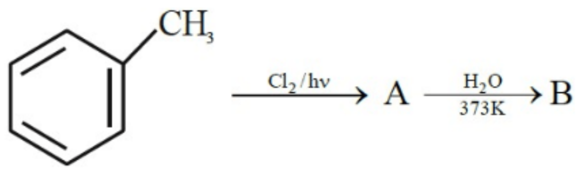
C. (A) is not correct but (R) is correct

D. Both (A) and (R) are correct and (R) is the correct explanation of (A)

Answer: A

Question48

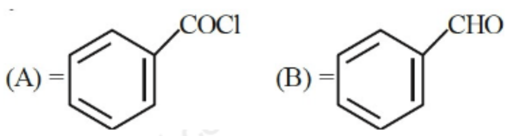
Identify A and B in the following sequence of reaction



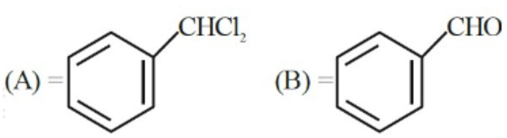
[1-Feb-2024 Shift 1]

Options:

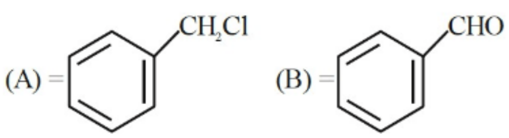
A.



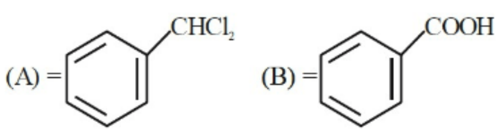
B.



C.



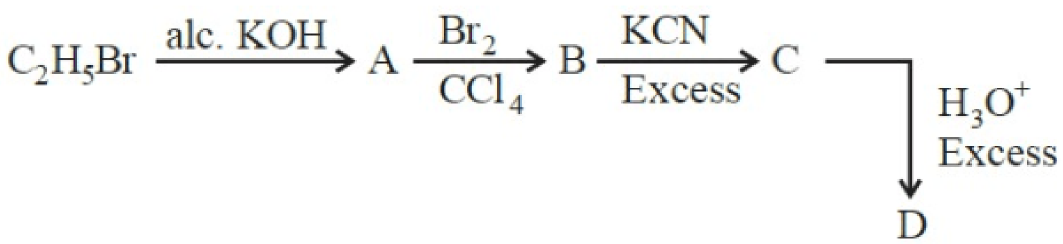
D.



Answer: B

Question49

Acid D formed in above reaction is :



[1-Feb-2024 Shift 2]

Options:

A. Gluconic acid

B. Succinic acid

C. Oxalic acid

D. Malonic acid

Answer: B

Question50

Assertion A : Hydrolysis of an alkyl chloride is a slow reaction but in the presence of NaI, the rate of the hydrolysis increases.
Reason R: I^- is a good nucleophile as well as a good leaving group.

In the light of the above statements, choose the correct answer from the options given below.
[24-Jan-2023 Shift 1]

Options:

A. A is false but R is true

B. A is true but R is false

C. Both A and R are true and R is the correct explanation of A

D. Both A and R are true but R is NOT the correct explanation of A

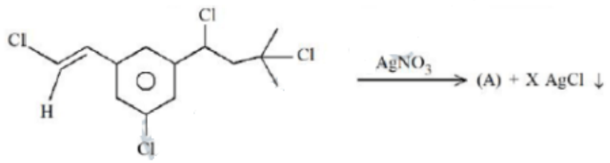
Answer: C

Solution:

The rate of hydrolysis of alkyl chloride improves because of better Nucleophilicity of I^-.

Question51

Number of moles of AgCl formed in the following reaction is



[24-Jan-2023 Shift 1]

Answer: 2

Solution:

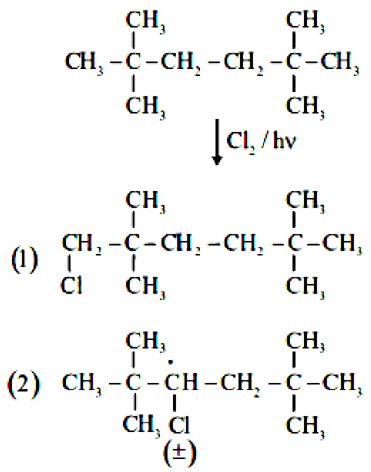
Benzylic and tertiary carbocations are stable.

Question52

Maximum number of isomeric monochloro derivatives which can be obtained from 2,2,5,5tetramethylhexane by chlorination is
[24-Jan-2023 Shift 2]

Answer: 3

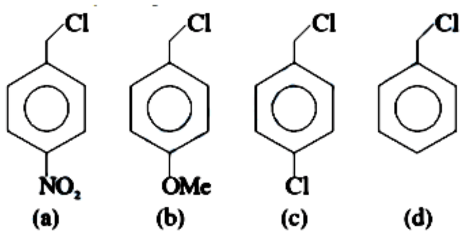
Solution:



Total numbers of isomer = 03

Question53

Decreasing order towards $S_N 1$ reaction for the following compounds is:



[30-Jan-2023 Shift 2]

Options:

A. a > c > d > b

B. a > b > c > d

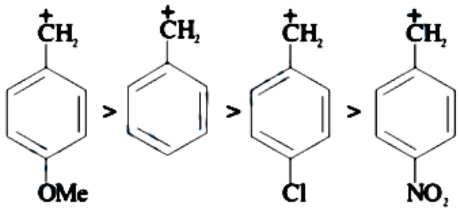
C. b > d > c > a

D. d > b > c > a

Answer: C

Solution:

The rate of S_N1 reaction depends upon stability of carbocation which follows the order



∴ Reactivity order

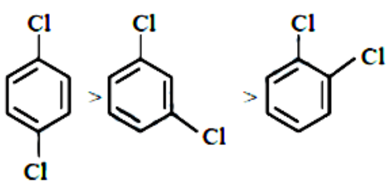
(b) > (d) > (c) > (a)

Question54

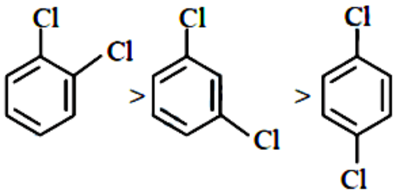
The correct order of melting point of dichlorobenzenes is [31-Jan-2023 Shift 1]

Options:

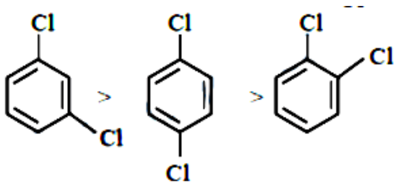
A.



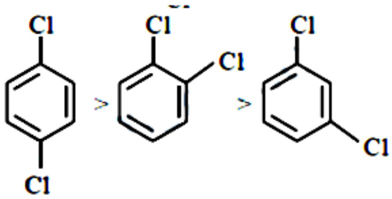
B.



C.

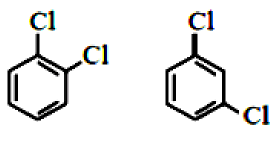


D.



Answer: D

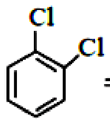
Solution:



b.p / K	453	446	448
m.p / K	256	249	323

 $\Rightarrow$ More Symmetrical & better Crystal fitting

M.P a Packing efficiency

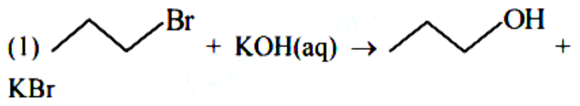
 $\Rightarrow \mu_{max}$

Question55

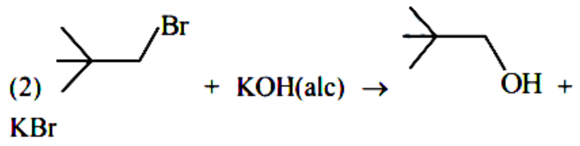
Identify the incorrect option from the following: [1-Feb-2023 Shift 1]

Options:

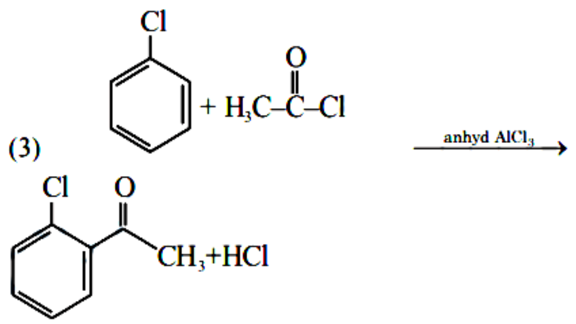
A.



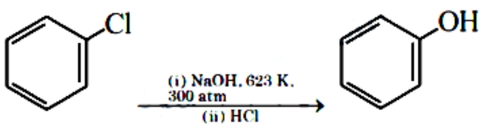
B.



C.



D.



Answer: B

Solution:

In alcoholic KOH, elimination reaction takes place.

Question56

For the reaction:

$$RCH_2Br + I^- \xrightarrow{\text{Acetone}} \underset{\text{major}}{RCH_2I} + Br^-$$

The correct statement is :
[6-Apr-2023 shift 1]

Options:

A. The transition state formed in the above reaction is less polar than the localised anion.

B. The reaction can occur in acetic acid also.

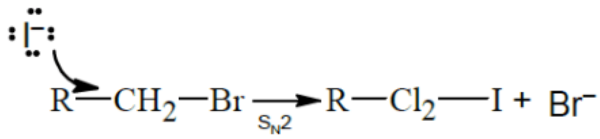
C. The solvent used in the reaction solvates the ions formed in rate determining step.

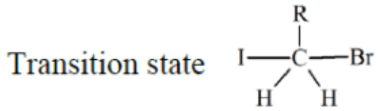
D. Br^- can act as competing nucleophile.

Answer: A

Solution:

This is finkelstein reaction





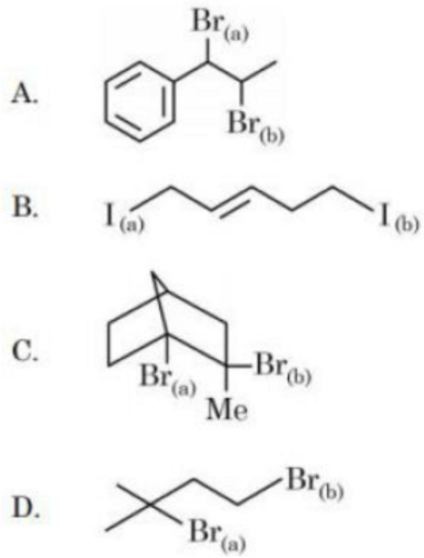
Clearly, the transition state is less polar than free anions. Br^{-} and I^{-}

Acetic acid is protic which does not support S_N2 Acetone does not solvate anion Br^{-} gets precipitated and hence can not compete with I^{-}

So only (1)is correct

Question57

Choose the halogen which is most reactive towards S_N1 reaction in the given compounds (A, B, C, & D)



[8-Apr-2023 shift 1]

Options:

A. $A-Br_{(a)}$; $B-I_{(a)}$; $C-Br_{(b)}$; $D-Br_{(a)}$

B. $A-Br_{(b)}$; $B-I_{(a)}$; $C-Br_{(a)}$; $D-Br_{(a)}$

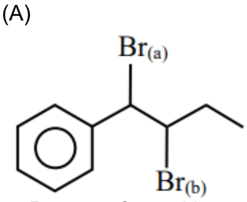
C. $A-Br_{(b)}$; $B-I_{(b)}$; $C-Br_{(b)}$; $D-Br_{(b)}$

D. $A-Br_{(a)}$; $B-I_{(a)}$; $C-Br_{(a)}$; $D-Br_{(a)}$

Answer: A

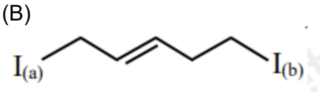
Solution:

(A)



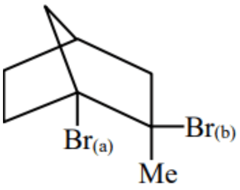
→ Because formed intermediate carbocation formed by $Br_{(a)}$ get stabilised by conjugation with phenyl

(B)



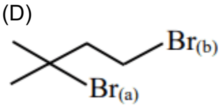
→ Because the intermediate carbocation formed by $I_{(a)}$ become more stable by conjugation

(C)



→ Because, we con't remove $Br_{(a)}$ from bridge head carbon (Bredt's rule)

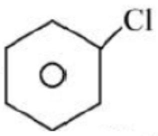
(D)

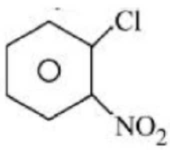


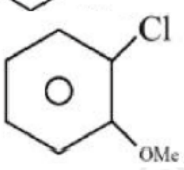
→ Because, formed intermediate by $Br_{(a)}$, 3° carbocation is more stable (stability of carbocation $3^{\circ} > 2^{\circ} > 1^{\circ}$)

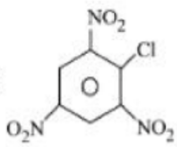
Question58

The correct order of reactivity of following haloarenes towards nucleophilic substitution with aqueous NaOH is

A. 

B. 

C. 

D. 

Choose the correct answer from the options given below:
[8-Apr-2023 shift 2]

Options:

A. D > B > A > C

B. A > B > D > C

C. C > A > D > B

D. D > C > B > A

Answer: A

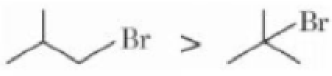
Solution:

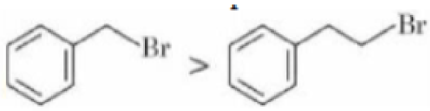
Rate α EWG α $\frac{1}{EDG}$

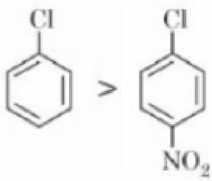
$NO_2 \rightarrow -$ Meffect

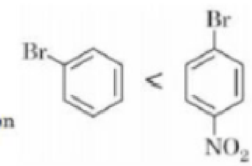
$OMe \rightarrow +M$ effect

Question59

(A) S_N2 

(B) S_N1 

(C) Electrophilic substitution 

(D) Nucleophilic substitution 

Choose the correct answer from the options given below:
[10-Apr-2023 shift 1]

Options:

A. (A), (C) and (D) only

B. (A), (B) and (D) only

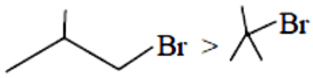
C. (B), (C) and (D) only

D. (A), (B), (C) and (D)

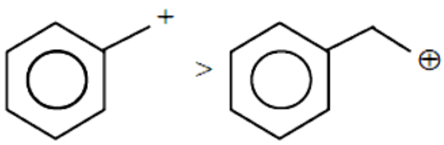
Answer: D

Solution:

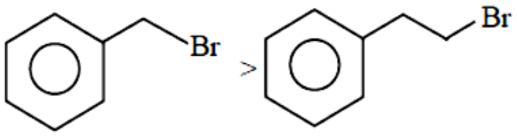
(A) $S_N2 \rightarrow$ for S_N2 Reaction $1^\circ > 2^\circ > 3^\circ$



(B) $S_N1 \rightarrow$ reactivity × Stability of Carbocation formed



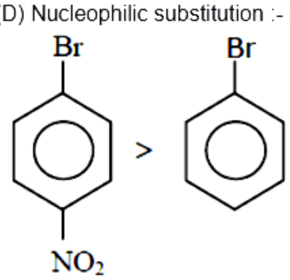
So,



(C) Electrophilic Substitution reaction

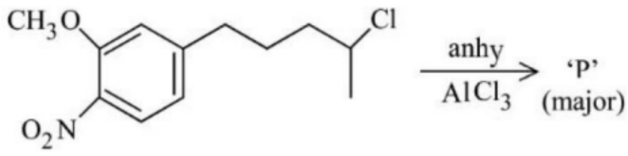
$$\text{rate} \times \frac{1}{EWG}$$

(D) Nucleophilic substitution :- rate × no. of EWG atloched at benzons



Question60

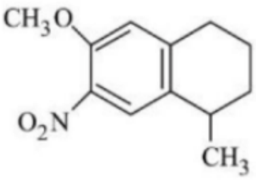
The major product ' P ' formed in the given reaction is:



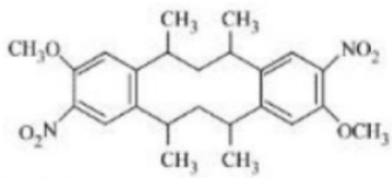
[10-Apr-2023 shift 2]

Options:

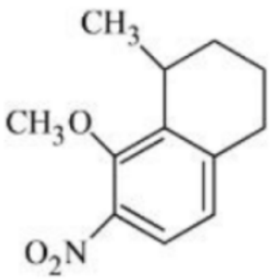
A.



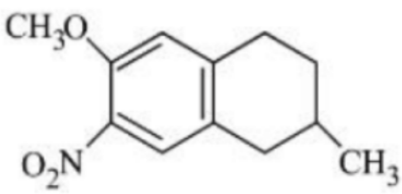
B.



C.

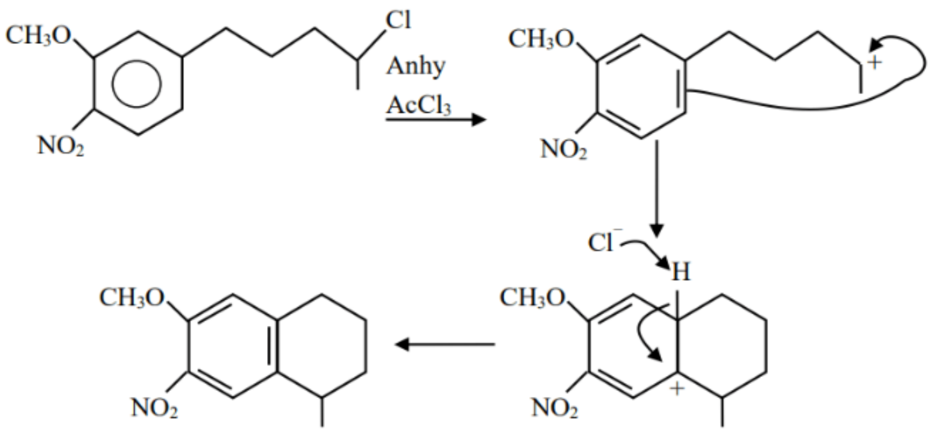


D.



Answer: A

Solution:



Question61

2-Methyl propyl bromide reacts with $C_2H_5O^-$ and gives ' A ' whereas on reaction with C_2H_5OH it gives ' B '. The mechanism followed in these reactions and the products ' A ' and ' B ' respectively are :
[13-Apr-2023 shift 1]

Options:

A. S_N1, A = tert-butyl ethyl ether; S_N1, B = 2-butyl ethyl ether

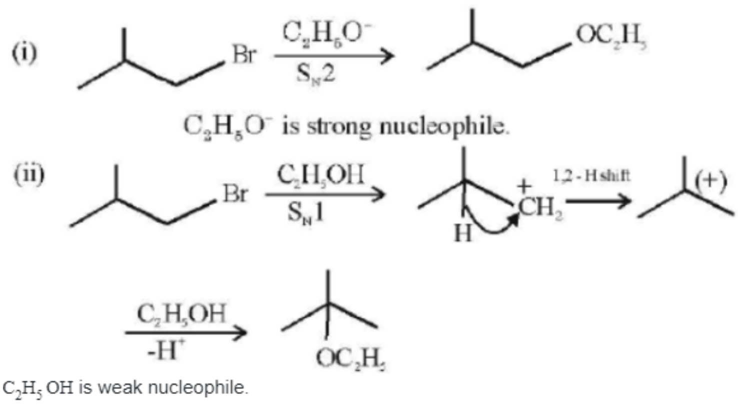
B. S_N2, A = 2-butyl ethyl ether; S_N2, B = iso-butyl ethyl ether

C. S_N, A = iso-butyl ethyl ether; S_N1, B = tert-butyl ethyl ether

D. S_N1, A = tert-butyl ethyl ether; S_N2, B = iso-butyl ethyl ether

Answer: C

Solution:



Question62

Match List I with List II
I - Bromopropane is reacted with reagents in List I to give product in List II

LIST I-Reagent	LIST II - Product
A KOH(alc)	I. Nitrile
B. KCN (alc)	II. Ester
C. $AgNO_2$	III. Alkene
D. $H_3CCOOAg$	IV. Nitroalkane

Choose the correct answer from the options given below :
[13-Apr-2023 shift 2]

Options:

A. A-IV, B-III, C-II, D-I

B. A-I, B-III, C-IV, D-II

C. A-I, B-II, C-III, D-IV

D. A-III, B-I, C-IV, D-II

Answer: D

Solution:

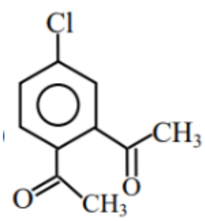
$$CH_3-CH_2-CH_2-Br+KOH(Alc) \rightarrow CH_3-\underset{(Allene)}{CH}=CH_2$$
$$CH_3-CH_2-CH_2-Br+KCN(Alc) \rightarrow CH_3-\underset{(Nitrle)}{CH_2}-CH_2-CN$$
$$CH_3-CH_2-CH_2-Br+AgNO+CH_3-\underset{(Niroallkane)}{CH_2}-CH_2-NO_2+AgBr\downarrow$$
$$CH_3-CH_2-CH_2-Br+CH_3-COOAg \rightarrow CH_3-COO-CH_2-\underset{(Ester)}{CH_2}-CH_3+AgBr\downarrow$$

Question63

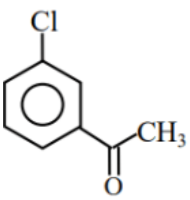
The major product in the Friedel-Craft acylation of chlorobenzene is :
[15-Apr-2023 shift 1]

Options:

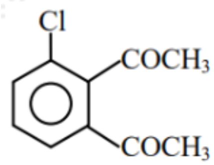
A.



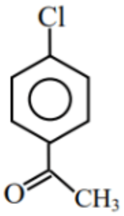
B.



C.

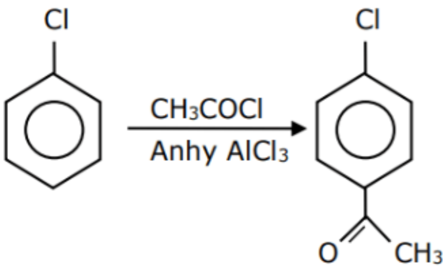


D.



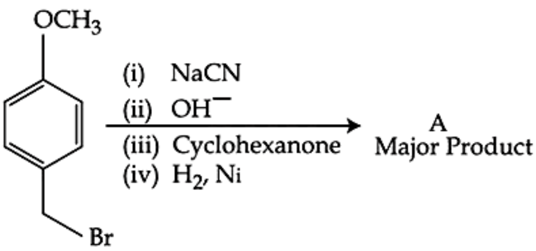
Answer: D

Solution:



Chlorine is ortho/para directing, para is major.

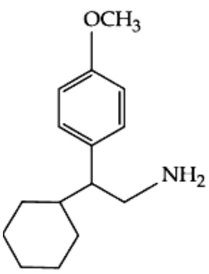
Question64



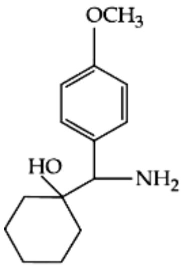
[24-Jun-2022-Shift-1]

Options:

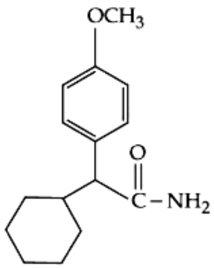
A.



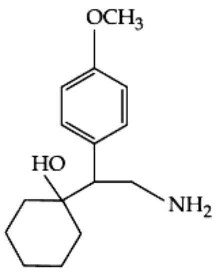
B.



C.

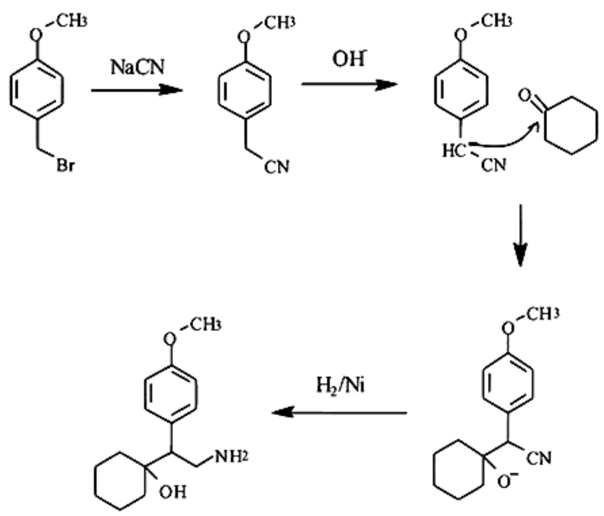


D.

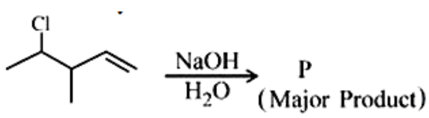


Answer: D

Solution:



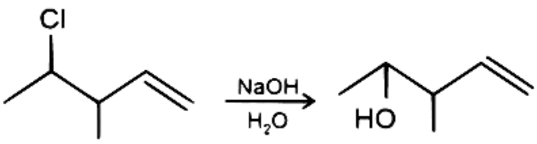
Question65



Consider the above reaction. The number of pi electrons present in the product ' P ' is_____ [24-Jun-2022-Shift-2]

Answer: 2

Solution:



P (Major product)

The given reaction undergoes nucleophilic substitution by SN2 mechanism at room temperature
∴ No. of π electrons present in $P = 2$

Question66

The major product in the reaction

$$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-Cl+\overset{+}{K}\ \overset{-}{O}-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-CH_3 \longrightarrow ?\ \text{is :}$$

[25-Jun-2022-Shift-1]

Options:

A. t-Butyl ethyl ether

B. 2,2-Dimethyl butane

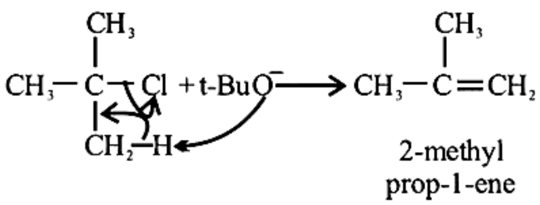
C. 2-Methyl pent-1-ene

D. 2-Methyl prop-1-ene

Answer: D

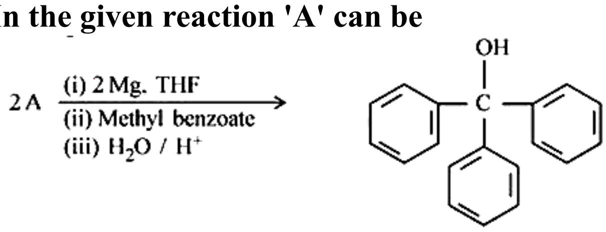
Solution:

We have been given a bulky base, hence elimination will take place & not substitution.



Question67

In the given reaction 'A' can be



[25-Jun-2022-Shift-2]

Options:

A. benzyl bromide

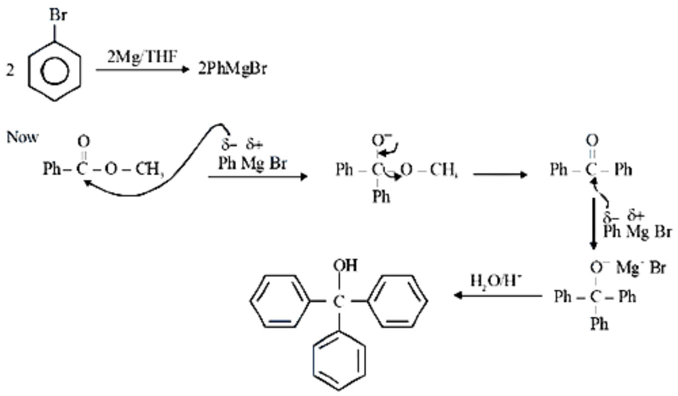
B. bromobenzene

C. cyclohexyl bromide

D. methyl bromide

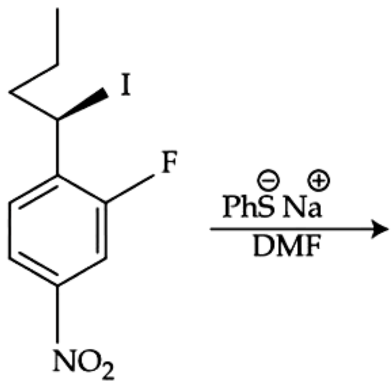
Answer: B

Solution:



Question68

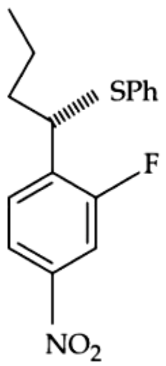
The major product of the following reaction is :



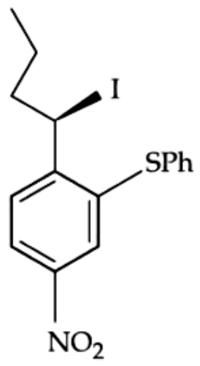
[27-Jun-2022-Shift-1]

Options:

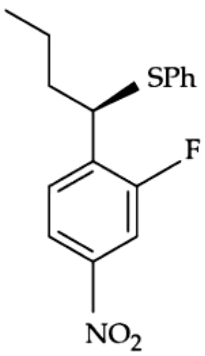
A.



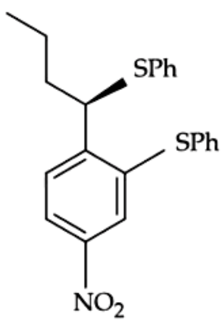
B.



C.

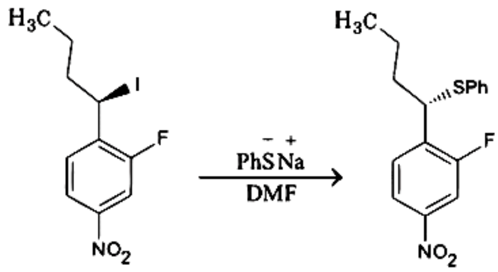


D.



Answer: A

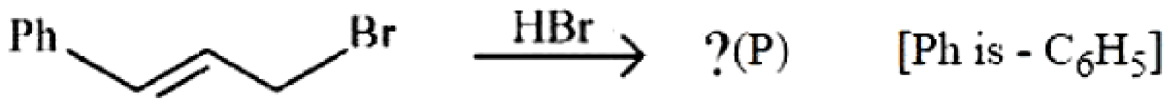
Solution:



It is bimolecular nucleophilic substitution (SN^2) which occur at benzylic carbon by inversion in configuration. This reaction cannot undergo substitution at benzene ring.

Question69

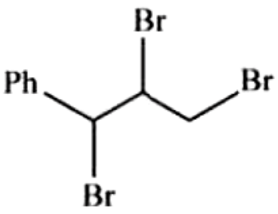
The major product (P) in the reaction



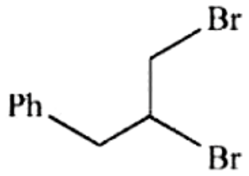
[28-Jun-2022-Shift-1]

Options:

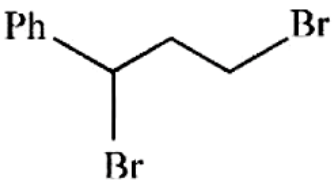
A.



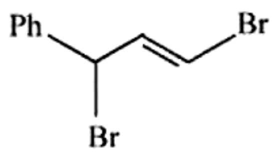
B.



C.

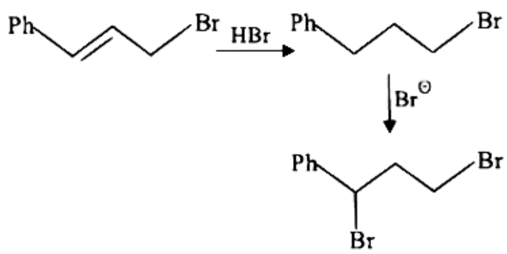


D.



Answer: C

Solution:



Question70

The correct structure of product 'A' formed in the following reaction.

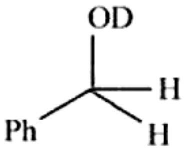
$$PhCHO + PH.CHO \xrightarrow[\text{in } D_2O]{NaOD} A + Ph-\overset{O}{\overset{\|}{C}}-O^{-} \text{(Ph is} - C_6H_5)$$

is

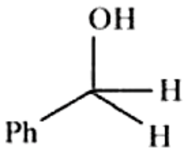
[28-Jun-2022-Shift-1]

Options:

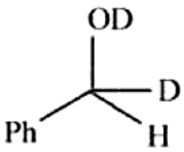
A.



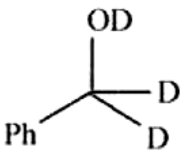
B.



C.

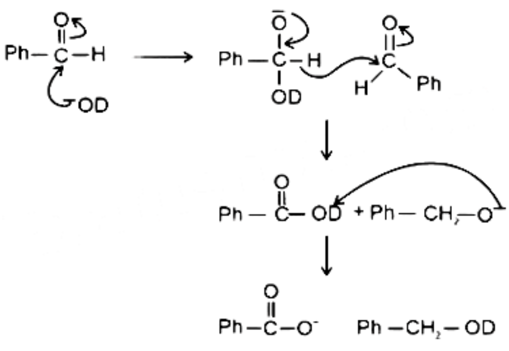


D.



Answer: A

Solution:



Question71

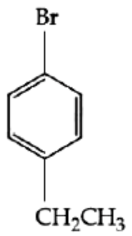
In the above reaction 'A' is

$$A \xrightarrow[\text{(ii) } CN^{-} \;\; \text{(iii) } H_2O/H^{+}]{\text{(i) } Cl_2, \Delta} \text{4-Bromophenyl acetic acid.}$$

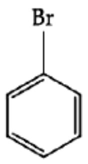
[28-Jun-2022-Shift-2]

Options:

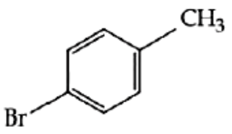
A.



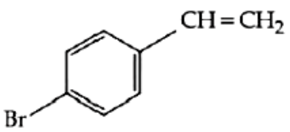
B.



C.

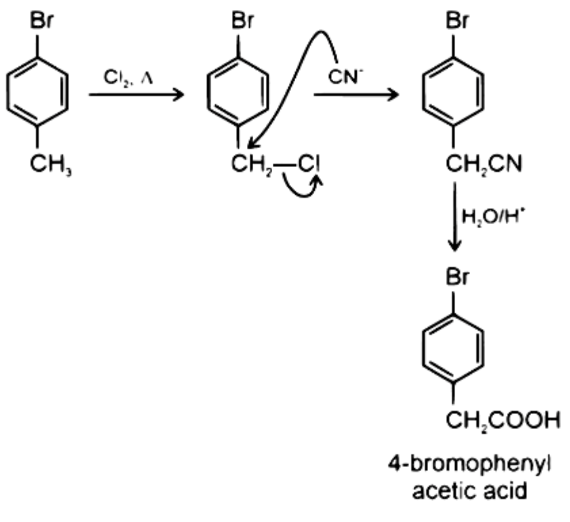


D.



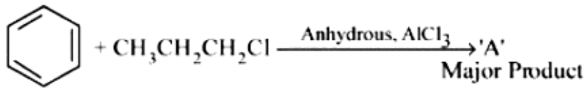
Answer: C

Solution:



Question72

The stable carbocation formed in the above reaction is



[29-Jun-2022-Shift-2]

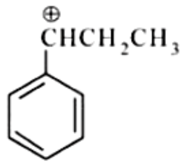
Options:

A. $CH_3CH_2\overset{\oplus}{C}H_2$

B. $CH_3\overset{\oplus}{C}H_2$

C. $CH_3-\overset{\oplus}{CH}-CH_3$

D.



Answer: C

Solution:

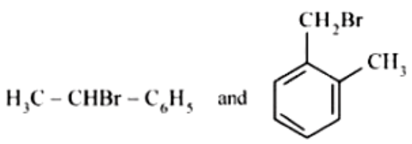
Initially $CH_3-CH_2-CH_2^+$ is formed. On rearrangement $CH_3-CH-CH_3$ stable carbocation is formed.

Question73

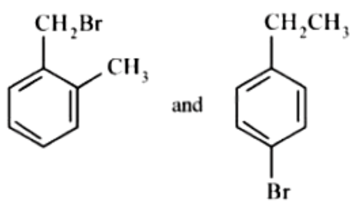
Two isomers (A) and (B) with Molar mass 184g/mol and elemental composition C, 52.2%; H, 4.9% and Br 42.9% gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with $KMnO_4$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $AgNO_3$. Isomer ' A ' and ' B ' are, respectively
[29-Jun-2022-Shift-2]

Options:

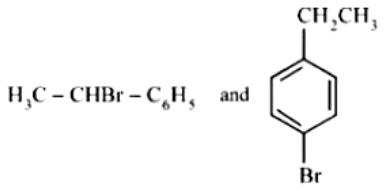
A.



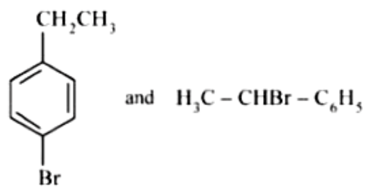
B.



C.

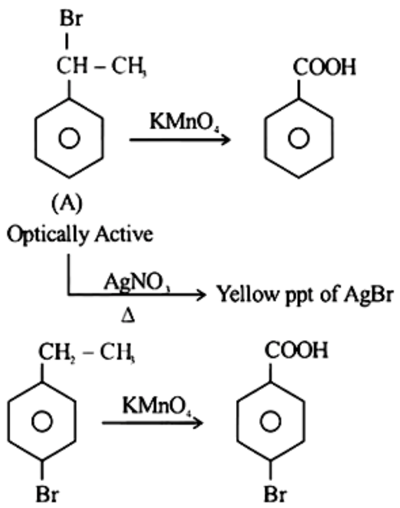


D.

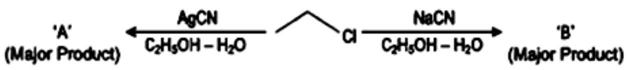


Answer: C

Solution:



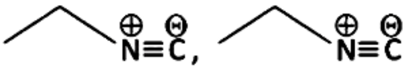
Question74



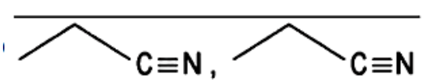
Considering the above reactions, the compound ' A′ and compound ' B ' respectively are : [29-Jul-2022-Shift-1]

Options:

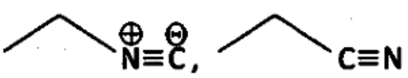
A.



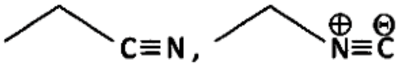
B.



C.

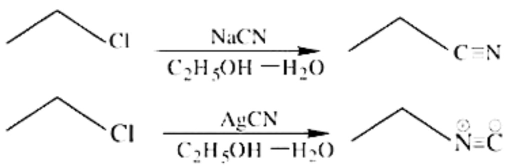


D.



Answer: C

Solution:

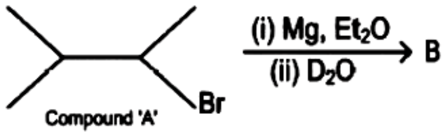


In NaCN; carbon is more nucleophilic atom.

Whereas in AgCN; Ag − C has covalent bond.

Question75

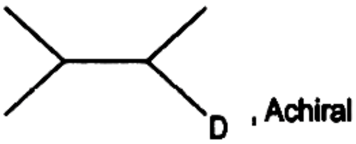
Compound 'A' undergoes following sequence of reactions to give compound ' B '. The correct structure and chirality of compound ' B ' is [where Et is $-C_2H_5$]



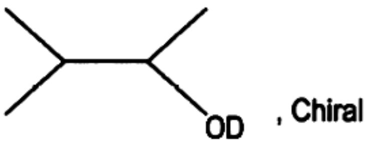
[29-Jul-2022-Shift-2]

Options:

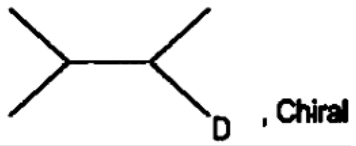
A.



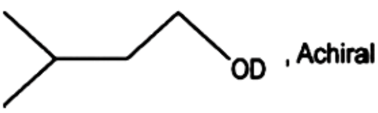
B.



C.

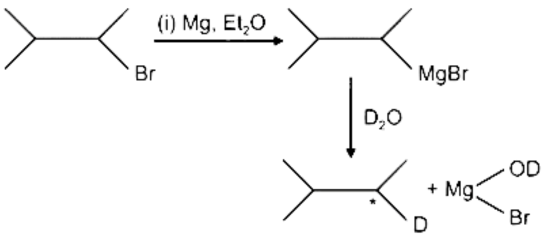


D.



Answer: C

Solution:



Question76

The product formed in the first step of the reaction of

$$\begin{array}{c} \quad\quad\quad\quad\quad\;\;\; Br \quad\quad\quad\quad\quad\quad\quad\\ CH_3-CH_2-\overset{|}{C}H-CH_2-\underset{|}{C}H-CH_3 \\ \quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\;\; Br \quad\; \end{array}$$

with excess Mg/Et_2O ($Et = C_2H_5$) text
[24 Feb 2021 Shift 1]

Options:

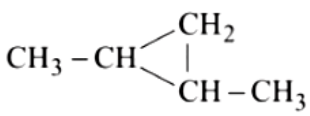
A.

$$\begin{array}{l} CH_3-CH_2-\underset{|}{C}H-CH_2-\underset{|}{C}H-CH_3 \\ \quad\quad\quad CH_3-CH-CH_2-CH-CH_2-CH_3 \end{array}$$

B.

$$\begin{array}{l} CH_2-CH_2-\underset{|}{C}H-CH_2-\underset{|}{C}H-CH_2 \\ CH_3-CH_2-CH-CH_2-CH-CH_3 \end{array}$$

C.

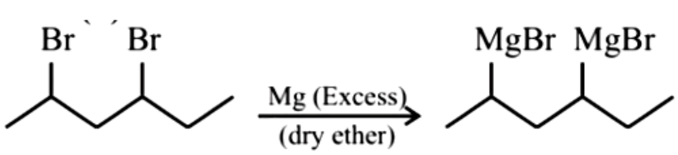


D.

$$\begin{array}{c} \quad\quad\quad MgBr \quad\quad\quad\quad\quad\quad\\ CH_3CH_2-\overset{|}{C}H-CH_2-\underset{|}{C}H-CH_3 \\ \quad\quad\quad\quad\quad\quad\quad\quad\quad MgBr \end{array}$$

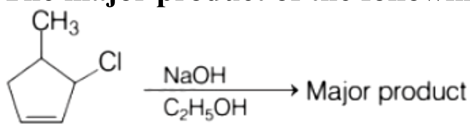
Answer: D

Solution:



Question77

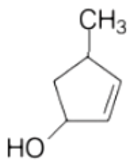
The major product of the following reaction is



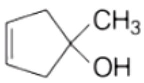
[31 Aug 2021 Shift 2]

Options:

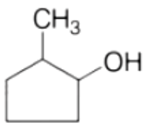
A.



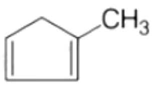
B.



C.



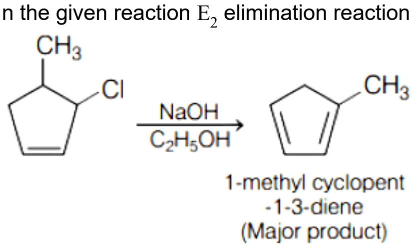
D.



Answer: D

Solution:

In the given reaction E_2 elimination reaction takes place in which two substituents are removed from a molecule to form double bond (alkene).

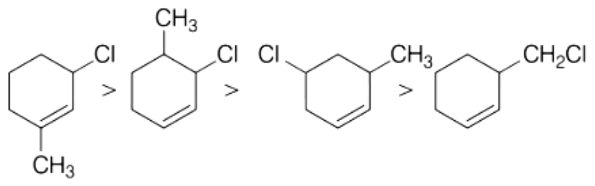


Question78

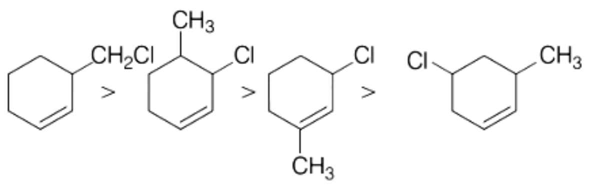
The correct order of reactivity of the given chlorides with acetate in acetic acid is
[31 Aug 2021 Shift 1]

Options:

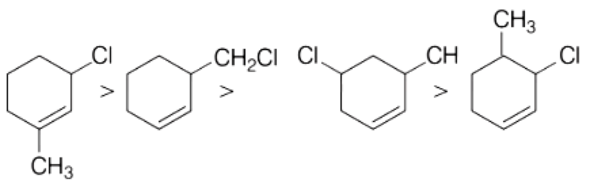
A.



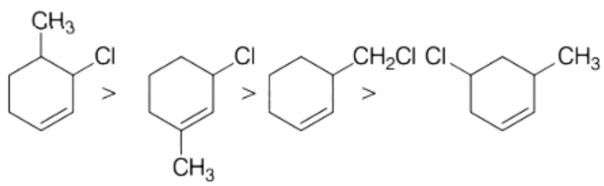
B.



C.



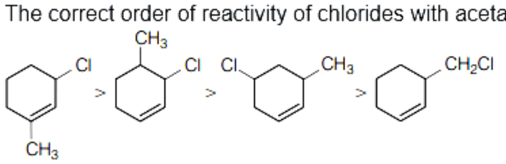
D.



Answer: A

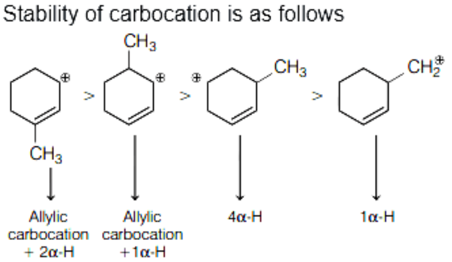
Solution:

The correct order of reactivity of chlorides with acetate in acetic acid is



The given chlorides undergoes S_N1 reaction. So, as the stability of carbocation formed increases, rate of reaction increases.

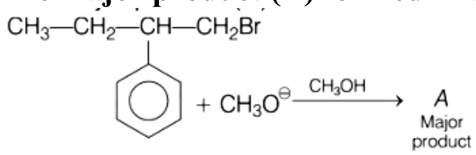
Stability of carbocation is as follows



Hence, correct option is (a).

Question79

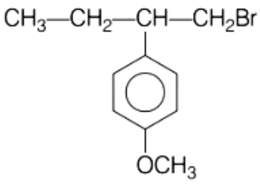
The major product (A) formed in the reaction given below is



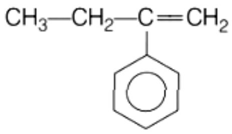
[27 Aug 2021 Shift 2]

Options:

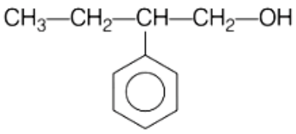
A.



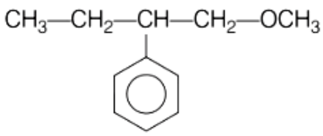
B.



C.



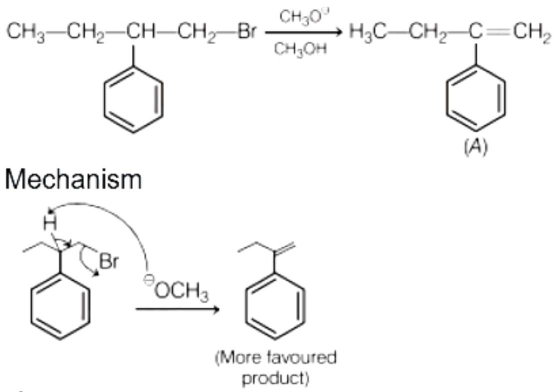
D.



Answer: B

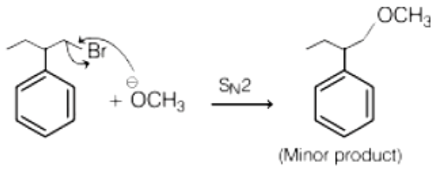
Solution:

On reaction of haloalkane with methoxide ion, alkene is formed.



● $\overset{\ominus}{O}CH_3$ will act as a base due to its small size and high electron density and therefore, abstracts proton to form double bond which is in conjugation with aromatic ring.

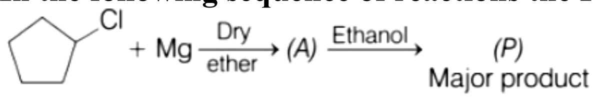
● The $\overset{\ominus}{O}CH_3$ when acts as nucleophile undergoes nucleophilic substitution and replaces $Br^{\ominus}$ to form ether, which is a minor product.



So, option (b) is correct.

Question80

In the following sequence of reactions the P is

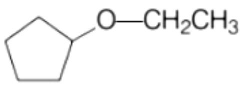


[27 Aug 2021 Shift 1]

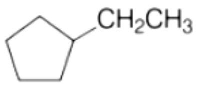
Options:

A.

B.



C.

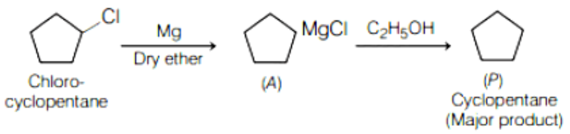


D.

Answer: A

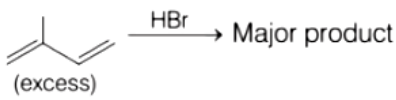
Solution:

In the first step, Grignard reagent is obtained. In the second step, acid-base reaction occurs. Due to presence of an acidic hydrogen in alcohol, neutralisation reaction takes place that produces alkane and water.
The reaction is



Question81

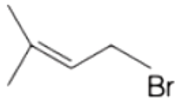
The major product formed in the following reaction is



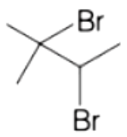
[26 Aug 2021 Shift 1]

Options:

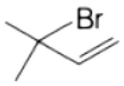
A.



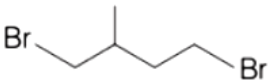
B.



C.

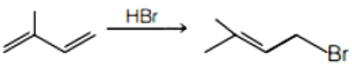


D.

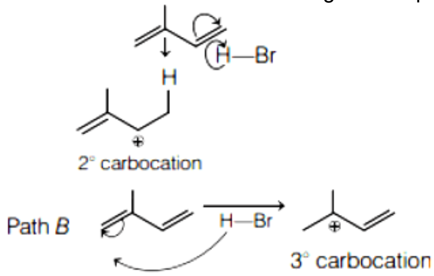


Answer: A

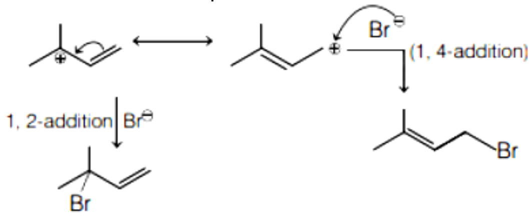
Solution:



2 - methylbut - 1, 3 - diene 1 - bromo 3 - methylbut - 2 - ene
This addition is known as 1, 4 - addition.
Mechanism This reaction can undergoes two pathways Path A



Out of the two intermediates formed in two paths, the path B intermediate is more stable as it has more stable carbocation. So, major product will be formed because of path B.



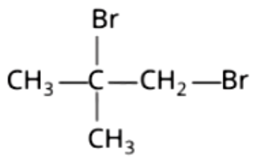
1, 2 -addition product is formed at low temperature and will be less stable as double bond is less substituted. 1, 4 - addition is thermodynamically stable product as double bond is more substituted. As diene is in excess and HBr is limited in reaction, so diene cannot be formed.
So option (b) is incorrect.

Question82

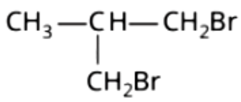
Excess of isobutane on reaction with Br_2 in presence of light at 125°C gives which one of the following, as the major product?
[26 Aug 2021 Shift 1]

Options:

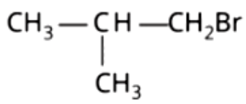
A.



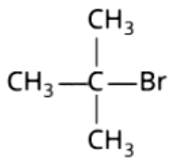
B.



C.



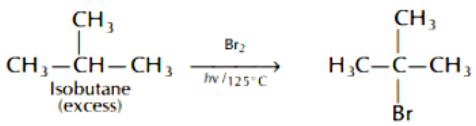
D.



Answer: D

Solution:

Excess of isobutane reacts with Br_2 in presence of light at $125°C$ gives 1 - bromo - 2 - methyl propane as major product.

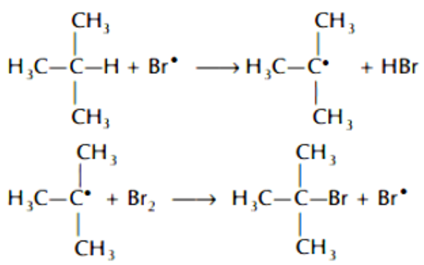


Mechanism The above reaction is halogenation of alkane via free radical substitution reaction.

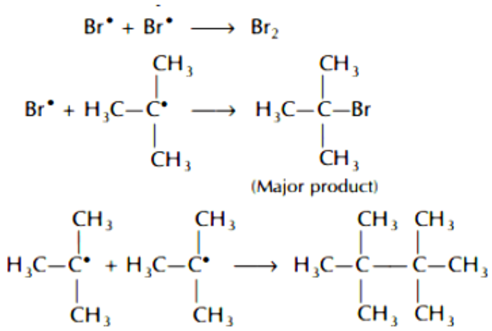
Initiation step

$Br_2 \xrightarrow{hv} Br^{\bullet} + Br^{\bullet}$

Propagation step



Termination step

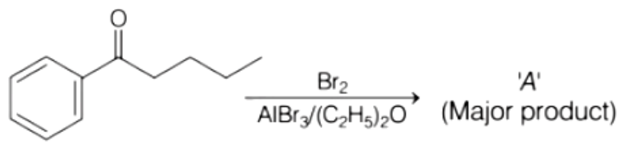


Reactivity order of abstraction of H towards bromination of alkane as more stable alkyl free radical is formed as follows

$3°H > 2°H > 1°H$

Since, isobutane is in excess, so dibromination of single isobutane is not favourable reaction. This make (a) and (b) incorrect options.

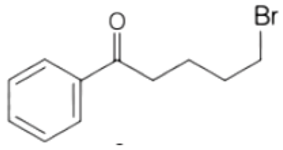
Question83



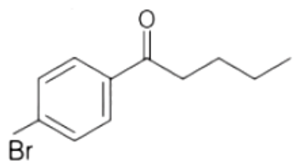
Consider the given reaction, the product A is
[26 Aug 2021 Shift 2]

Options:

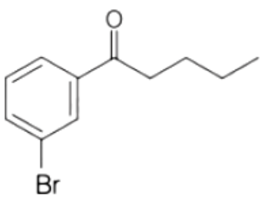
A.



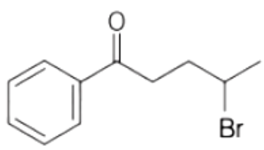
B.



C.



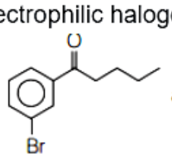
D.

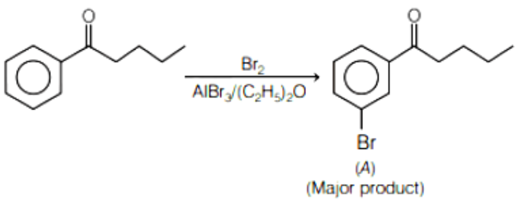


Answer: C

Solution:

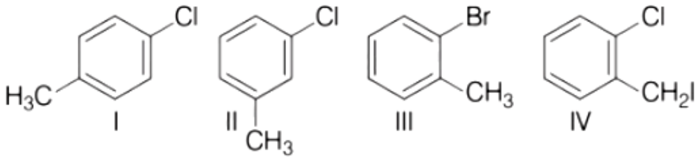
In presence of Lewis acid, electrophilic halogenation reaction

takes place at *meta* position to give  as follows



Question84

Among the following compounds I - IV, which one forms a yellow precipitate on reacting sequentially with
(i) NaOH
(ii) dil. HNO_3
(iii) $AgNO_3$?



[26 Aug 2021 Shift 1]

Options:

A. II

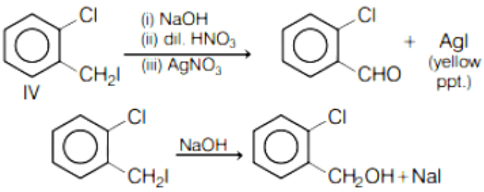
B. IV

C. I

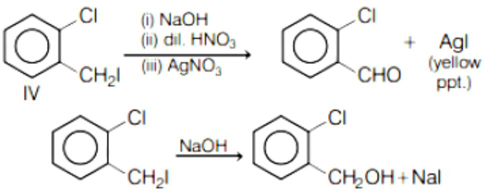
D. III

Answer: B

Solution:



This compound halide will only give yellow ppt. as benzyl carbocation formed shown below is highly stable by conjugation.



Other compounds halide cannot be removed because their corresponding carbocation is highly unstable.

Question85

In the following sequence of reactions,

$$C_3H_6 \xrightarrow{H^+/H_2O} A \xrightarrow[\text{dil. KOH}]{\text{KIO}} B + C$$

The compounds B and C respectively are
[1 Sep 2021 Shift 2]

Options:

A. Cl_3COOK, HCOOH

B. Cl_3COOK, CH_3I

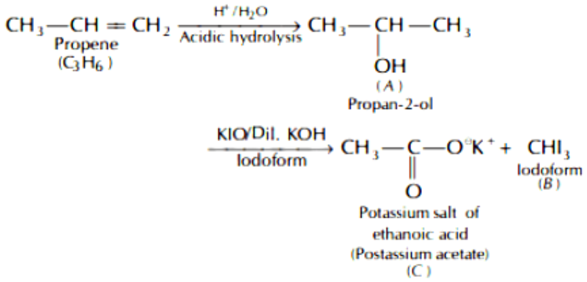
C. CH_3, HCOOK

D. CHI_3, CH_3COOK

Answer: D

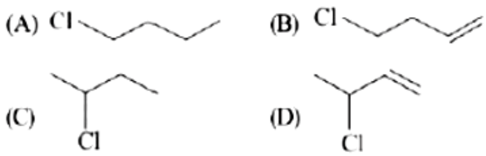
Solution:

Propene (C_3H_6) undergoes acidic hydrolysis to give A which is 2° alcohol. This alcohol undergoes iodoform reaction in presence of KIO and dil. KOH to give iodoform along with potassium salt of carboxylic acid.
This reaction is known as iodoform test.



Question86

The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is:



[Jan. 08,2020 (I)]

Options:

A. D > B > C > A

B. B > D > A > C

C. B > D > C > A

D. B > A > D > C

Answer: A

Solution:

E_1 reaction proceeds via carbocation formation, therefore greater the stability of carbocation, faster will be the E_1 reaction. Thus correct decreasing order of the given halides towards dehydrohalogenation by E_1 is
D > B > C > A

Question87

Consider the following reactions:

(1) $(CH_3)_3CCH(OH)CH_3 \xrightarrow{\text{conc. } H_2SO_4}$

(2) $(CH_3)_2CH\,CH(Br)CH_3 \xrightarrow{\text{alc KOH}}$

(3) $(CH_3)_2CH\,CH(Br)CH_3 \xrightarrow{(CH_3)_3O^-K^{\oplus}}$

(4) $-CH_2-CHO \xrightarrow{\Delta}$

Which of these reaction(s) will not produce Saytzeff product?
[Jan. 07,2020 (I)]

Options:

A. (1), (3) and (4)

B. (4) only

C. (3) only

D. (2) and (4)

Answer: C

Solution:

(A) $(CH_3)_3CCH(OH)CH_3 \xrightarrow{\text{conc. } H_2SO_4} CH_3-\underset{CH_3}{\overset{CH_3}{C}}=C-CH_3$

(B) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{\text{alc. KOH}} CH_3-\overset{CH_3}{C}=CH-CH_3$

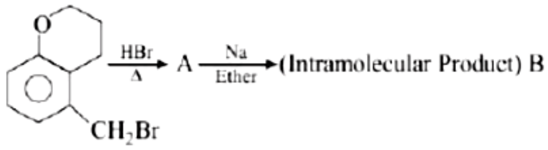
(C) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{1-BuO^-K^+} CH_3-\overset{CH_3}{CH}-CH=CH_2$

Due to bulky nature of tertiary butoxide, the least hindered hydrogen is eliminated. Therefore, Hoffiman product is formed.

(D) $(CH_3)_2\underset{OH}{C}-CH_2-CHO \xrightarrow{\Delta} CH_3-\overset{CH_3}{C}=CH-CHO$

Question88

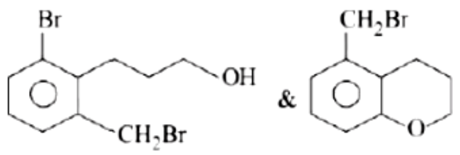
In the following reaction sequence, structures of A and B, respectively will be:



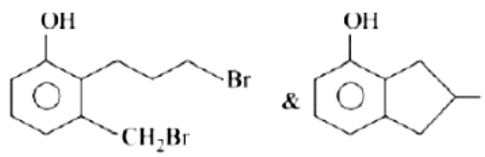
[Jan. 07, 2020 (II)]

Options:

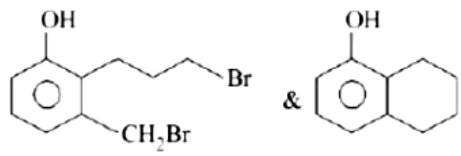
A.



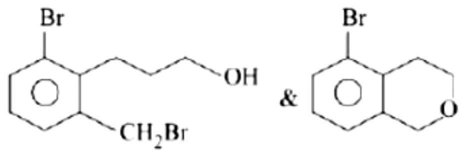
B.



C.

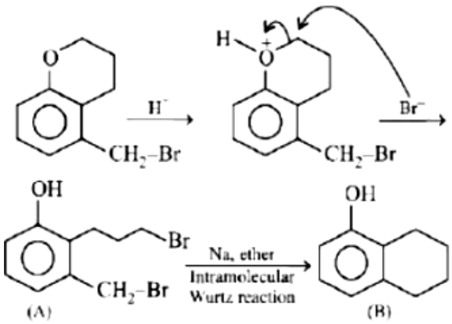


D.



Answer: C

Solution:

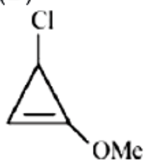


Question89

The decreasing order of reactivity of the following organic molecules towards $AgNO_3$, solution is:

(A)

(B)



(C) $CH_3\underset{Cl}{\underset{|}{C}}HCH_3$

(D) $CH_3\underset{Cl}{\underset{|}{C}}HCH_2NO_2$

[Sep. 04,2020 (I)]

Options:

A. (C) > (D) > (A) > (B)

B. (A) > (B) > (D) > (C)

C. (A) > (B) > (C) > (D)

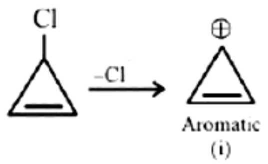
D. (B) > (A) > (C) > (D)

Answer: D

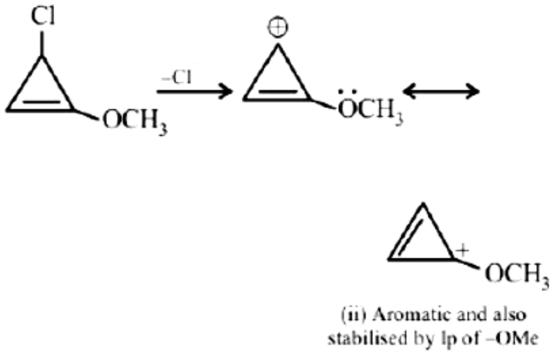
Solution:

Given reaction is S_N1 reaction. In S_N1 reaction Rate of reaction ∝ Stability of C^+

(A)



(B)



(C) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3 \xrightarrow{-Cl} CH_3-\underset{\substack{(2°)\\(iii)}}{\overset{\oplus}{C}H}-CH_3$

(D) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_2-NO_2 \xrightarrow{-Cl} CH_3-\underset{(iv)}{\overset{\oplus}{C}H}-CH_2-\underset{(-I)}{NO_2}$

Stability of C^+ : ii > i > iii > iv

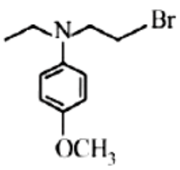
Reactivity order : B > A > C > D

Question90

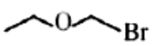
Which of the following compounds will form the precipitate with aq. $AgNO_3$ solution most readily? [Sep. 04,2020 (II)]

Options:

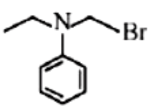
A.



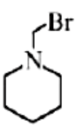
B.



C.



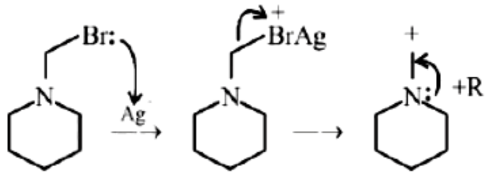
D.



Answer: D

Solution:

Ease of precipitation of AgBr depends upon the rate of formation of carbocation.



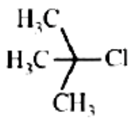
Most stable carbocation due to +R effect of N.

Question91

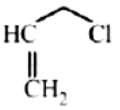
Among the following compounds, which one has the shortest C - Cl bond? [Sep. 04,2020 (II)]

Options:

A.

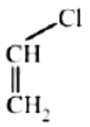


B.



C. H_3C-Cl

D.



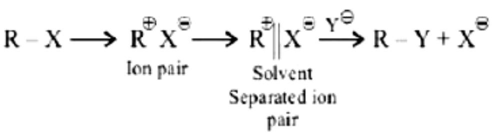
Answer: D

Solution:

Due to conjugation of lonepair of Cl with π bond, partial double bond character decreases bond length that's why compound (d) has shortest $C-Cl$ bond length.

Question92

The mechanism of S_N1 reaction is given as:



A student writes general characteristics based on the given mechanism as:
(1) The reaction is favoured by weak nucleophiles.
(2) $R^{\oplus}$ would be easily formed if the substituents are bulky.
(3) The reaction is accompanied byracemization.
(4) The reaction is favoured by non-polar solvents. Which observations are correct? [Sep. 03,2020 (I)]

Options:

A. (1) and (2)

B. (1) and (3)

C. (1), (2) and (3)

D. (2) and (4)

Answer: C

Solution:

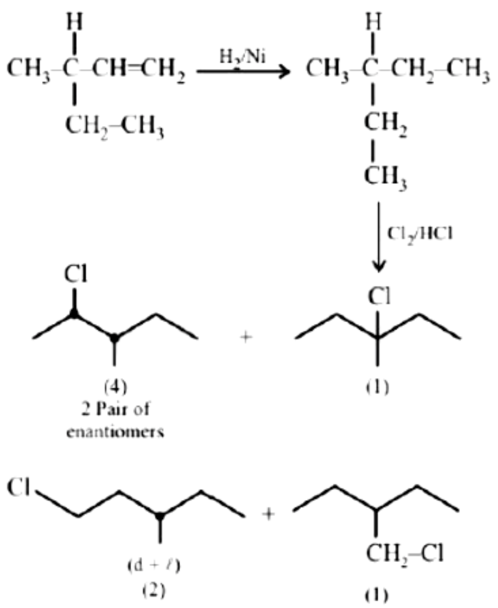
Above reaction is S_N1 reaction as it proceeds via formation of carbocation. Polar protic solvent is more suitable for S_N1 and so racemisation takes place.

Question93

The total number of monohalogenated organic products in the following (including stereoisomers) reaction is __________. $\underset{\text{(Simplest optically active alkene)}}{A} \xrightarrow[\text{(ii)}X_2/\Delta]{\text{(i)}H_2/Ni/\Delta}$
[NV, Sep. 03,2020 (I)]

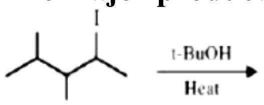
Answer: 8

Solution:



Question94

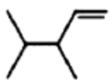
The major product in the following reaction is:



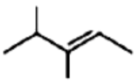
[Sep. 03,2020 (II)]

Options:

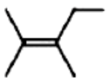
A.



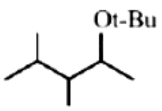
B.



C.

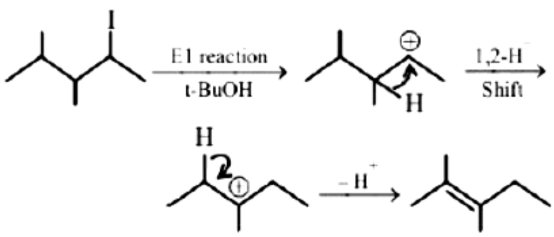


D.



Answer: C

Solution:



Question95

Which of the following compounds will show retention in configuration on nucleophic substitution by OH^- ion?
[Sep. 02,2020 (I)]

Options:

A. $CH_3-\overset{Br}{\underset{C_6H_{13}}{C}}-H$

B. $CH_3-\underset{C_6H_5}{CH}-Br$

C. $CH_3-\underset{CH_3}{CH}-Br$

D. $CH_3-\underset{\underset{C_2H_5}{|}}{CH}-CH_2Br$

Answer: D

Solution:

$$CH_3-\underset{\underset{C_2H_5}{|}}{CH}-CH_2Br \xrightarrow{S_N1} CH_3-\underset{\underset{C_2H_5}{|}}{CH}-\overset{+}{C}H_2 \xrightarrow{1.2H^{-}\text{ shift}} CH_3-\underset{\underset{C_2H_5}{|}}{\overset{+}{C}}-CH_3 \longrightarrow CH_3-\underset{\underset{C_2H_5}{|}}{\overset{\overset{OH}{|}}{C}}-CH_3$$

Question96

The major product obtained from E 2 -elimination of 3-bromo-2-fluoropentane is: [Sep. 02,2020 (II)]

Options:

A. $CH_3CH_2-\overset{\overset{Br}{|}}{CH}-CH=CH_2$

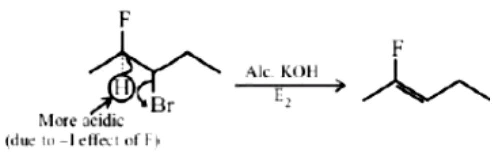
B. $CH_3CH_2CH=\underset{\underset{CH_3}{|}}{C}-F$

C. $CH_3-CH=CH-\overset{\overset{F}{|}}{CH}-CH_3$

D. $CH_3-CH_2-\overset{\overset{Br}{|}}{C}-CH=CH_3$

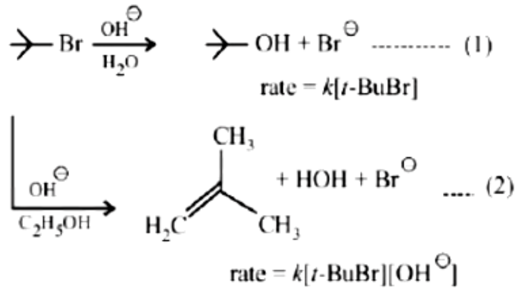
Answer: B

Solution:



Question97

Consider the reaction sequence given below:



Which of the following statements is true? [Sep. 02,2020 (II)]

Options:

A. Changing the base from $OH^{\ominus}$ to $^{\ominus}OR$ will have no effect on reaction (2).

B. Changing the concentration of base will have no effect on reaction (1).

C. Doubling the concentration of base will double the rate of both the reactions.

D. Changing the concentration of base will have no effect on reaction (2).

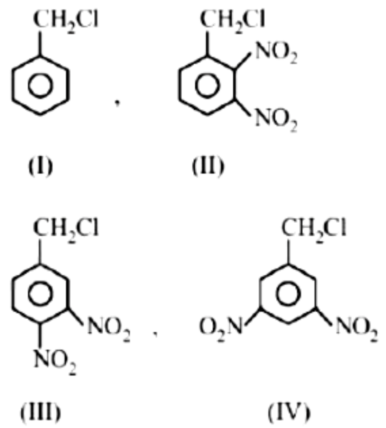
Answer: B

Solution:

First reaction is $S_N 1$ in which rate does not depend on conc. of nucleophile but depends on reactant conc. Second reaction is $E2$ reaction in which rate depends on conc. of base as well as reactant conc. Therefore, changing in the concentration of base will have no effect on rate of reaction (1).

Question98

The decreasing order of reactivity of the following compounds towards nucleophilic substitution ($S_N 2$) is :



[Sep. 03,2020 (II)]

Options:

A. $(II) > (III) > (I) > (IV)$

B. $(II) > (III) > (IV) > (T)$

C. $(III) > (II) > (IV) > (I)$

D. $(IV) > (II) > (III) > (I)$

Answer: B

Solution:

$S_N{}^2$ reactions depend upon $-I$ and $-M$ effect on substrate. On increasing $-I$ and $-M$ effect, rate of $S_N 2$ reaction will increase.

Question99

The major product of the following reaction is: $CH_3CH_2\underset{|\atop Br}{CH} - \underset{|\atop Br}{CH_2} \xrightarrow[\text{(ii) Na NH}_2\text{ in liq . NH}_2]{\text{(i) KOH atc}}$

[Jan. 12,2019(II)]

Options:

A. $CH_3CH = C = CH_2$

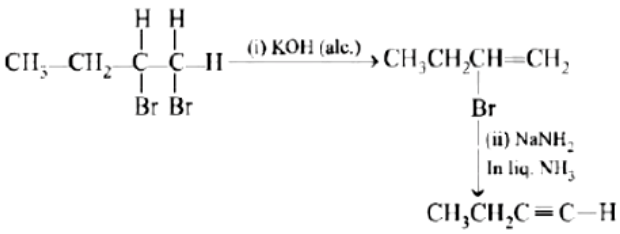
B. $CH_3CH_2\underset{NH_2}{CH}-\underset{NH_2}{CH_2}$

C. $CH_3CH=CHCH_2NH_2$

D. $CH_3CH_2C\equiv CH$

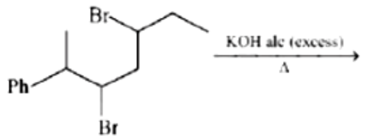
Answer: D

Solution:



Question100

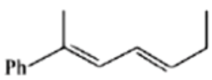
The major product of the following reaction is:



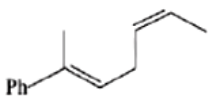
[Jan. 10,2019(I)]

Options:

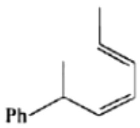
A.



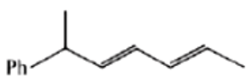
B.



C.



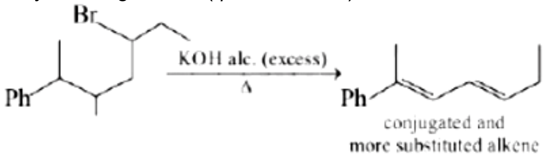
D.



Answer: A

Solution:

Dehydrohalogenation (β -elimination) occurs as:



Question101

Which hydrogen in compound (E) is easily replaceable during bromination reaction in presence of light?

$\underset{\delta}{CH_3} - \underset{\gamma}{CH_2} - \underset{\beta}{CH} = \underset{\alpha}{CH_2}$

[Jan. 10,2019(I)]

Options:

A. α - hydrogen

B. γ - hydrogen

C. δ - hydrogen

D. β - hydrogen

Answer: B

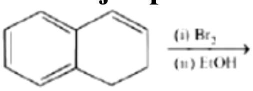
Solution:

Allylic H is easily replaced due to the greater stability of allylic free radical.

$$CH_3 - CH_2 - CH = CH_2 + Br_2 \xrightarrow{hv} CH_3 - \underset{|\atop Br}{CH} - CH = CH_2$$

Question102

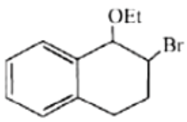
The major product of the following reaction is:



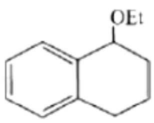
[Jan. 9, 2019(I)]

Options:

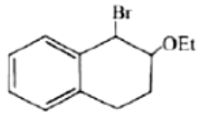
A.



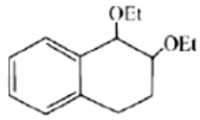
B.



C.



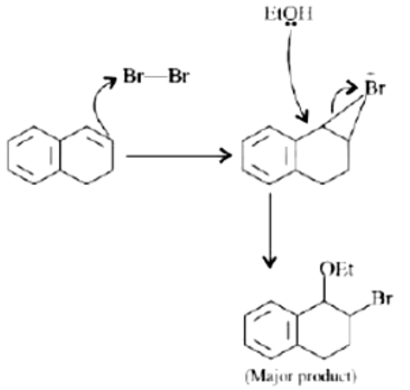
D.



Answer: A

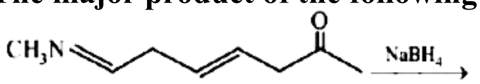
Solution:

Mechanism involved for the given reaction is:



Question103

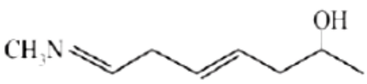
The major product of the following reaction is:



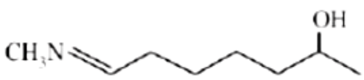
[Jan. 10,2019(II)]

Options:

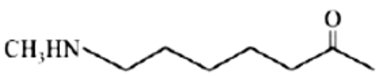
A.



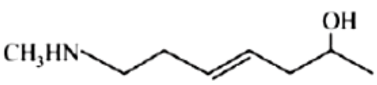
B.



C.



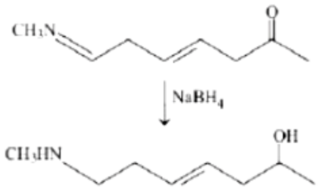
D.



Answer: D

Solution:

Sodium borohydride is a selective reducing agent. It reduces carbonyl group to alcoholic group, N methylimino group ($MeN = CH -$) to 2° amines, but does not reduce an isolated carbon-carbon double bond. Reaction involved:



(major product)

Question104

An 'Assertion' and a 'Reason' are given below. Choose the correct answer from the following options: Assertion (A): Vinyl halides do not undergo nucleophilic substitution easily. Reason (R): Even though the intermediate carbocation is stabilized by loosely held π -clectrons, the cleavage is difficult because of strong bonding.
[April 12, 2019 (II)]

Options:

A. Both (A) and (R) are wrong statements.

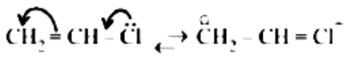
B. Both (A) and (R) are correct statements and (R) is the correct explanation of (A)

C. Both(A) and (R) are correct statements but (R) is not the correct explanation of (A).

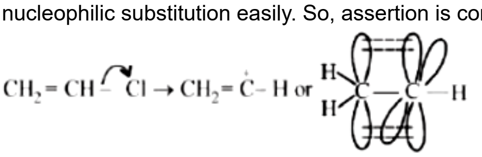
D. (A) is a correct statement but (R) is a wrong statement.

Answer: D

Solution:



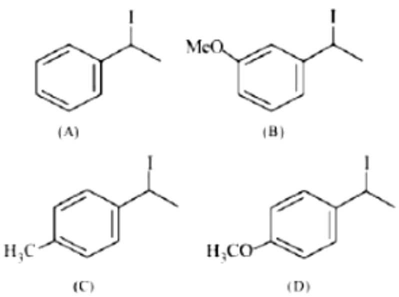
Due to partial double bond character of C -halogen bond, halogen leaves with great difficulty, if at all it does. Hence, vinyl halides do not undergo nucleophilic substitution easily. So, assertion is correct.



Intermediate carbocation is not stabilised by loosely heldπ electrons because empty orbital, being at 90°, cannot overlap with p -orbitals of π bond. So, reason is wrong.

Question105

Increasing rate of S_N1 reaction in the following compounds is :



[April 10, 2019 (I)]

Options:

A. (A) < (B) < (C) < (D)

B. (B) < (A) < (C) < (D)

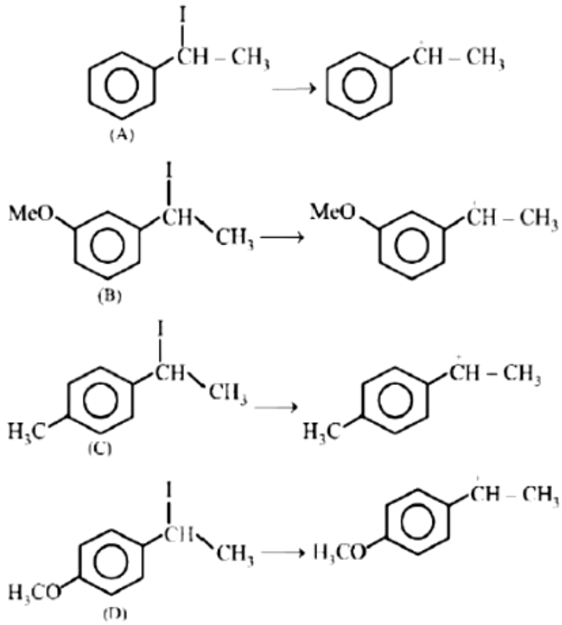
C. (B) < (A) < (D) < (B)

D. (A) < (B) < (D) < (C)

Answer: B

Solution:

The rate of $S_N 1$ is decided by the stability of carbocation formed in the rate determining step.



Carbocation(D) is most stable due to $+R$ effect of $-OCH_3$ group; (C) is stabilised by $+I$ and $+H$ effects of the CH_3 group; (B) is least stable due to $-$I effect of MeO group and (A) is stabilised by $-CH_3$ as well as phenyl group. So increasing order of rate of $S_N 1$ is

$(B) < (A) < (C) < (D)$

Question106

The major product of the following reaction is $CH_3 - \underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}} = \underset{\underset{Br}{|}}{CHCH_3} \xrightarrow{CH_3OH}$

[April 10, 2019 (I)]

Options:

A. $CH_3 - \underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}} = CH = CH_2$

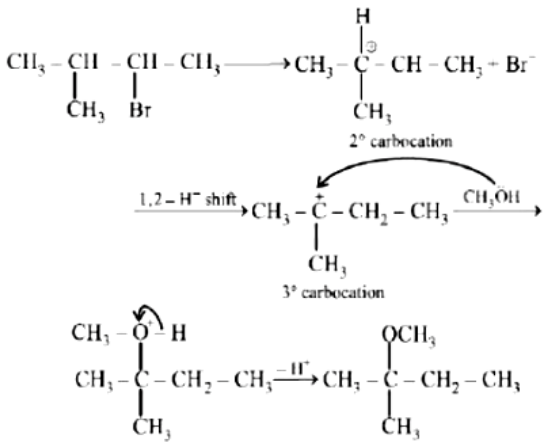
B. $CH_3 - \overset{\overset{CH_3}{|}}{C} = CHCH_3$

C. $CH_3 - \underset{\underset{OCH_3}{|}}{\overset{\overset{CH_3}{|}}{C}} = CH_2CH_3$

D. $CH_3 - \underset{\underset{H}{|}}{\overset{\overset{CH_3}{|}}{C}} = \underset{\underset{OCH_3}{|}}{CHCH_3}$

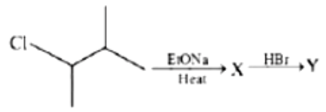
Answer: C

Solution:



Question107

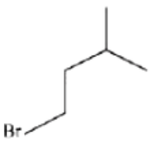
The major product ' Y ' in the following reaction is:



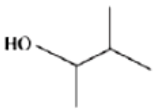
[April 10, 2019 (II)]

Options:

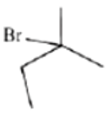
A.



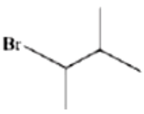
B.



C.

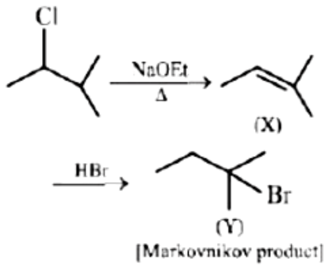


D.



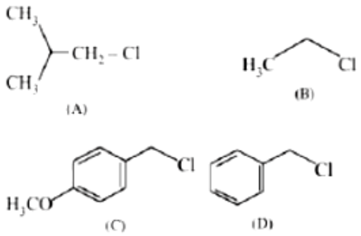
Answer: C

Solution:



Question108

Increasing order of reactivity of the following compounds for $S_N 1$ substitution is:



[April 9, 2019 (II)]

Options:

A. (B) < (C) < (D) < (A)

B. (B) < (C) < (A) < (D)

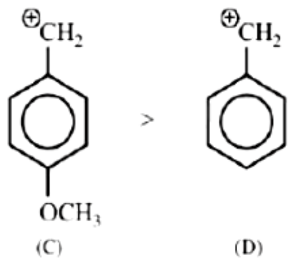
C. (B) < (A) < (D) < (C)

D. (A) < (B) < (D) < (C)

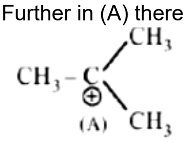
Answer: C

Solution:

In $S_N 1$ reaction carbocation acts as an intermediate.



Carbocation produced by (C) is more stable than carbocation produced by (D) due to $+I$ effect of $-OCH_3$ group.
Further in (A) there is formation of tertiary carbocation after rearrangement while (B) is primary carbocation.



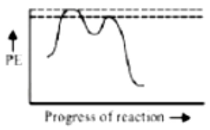
So, the correct order is (C) > (D) > (A) > (B)

Question109

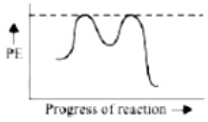
Which of the following potential energy (PE) diagrams represents the S_N I reaction? [April 9, 2019 (II)]

Options:

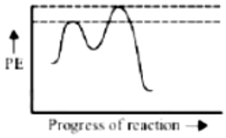
A.



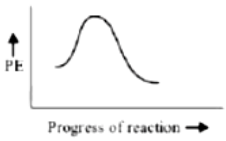
B.



C.



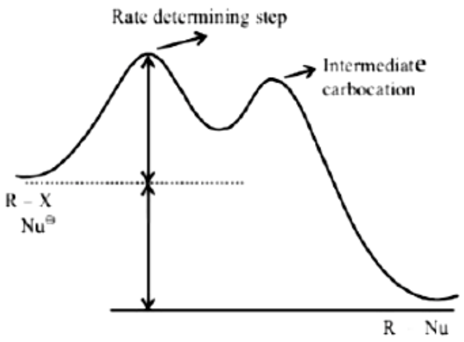
D.



Answer: A

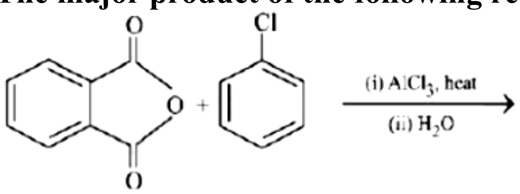
Solution:

The S_N1 reaction energy diagram illustrates the dominant part of the substrate with respect to the reaction rate. The rate determining step is the formation of the intermediate carbocation.



Question110

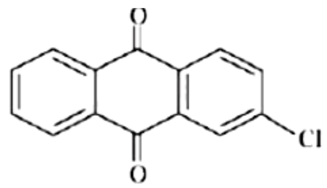
The major product of the following reaction is:



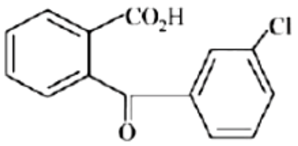
[April 8,2019 (1)]

Options:

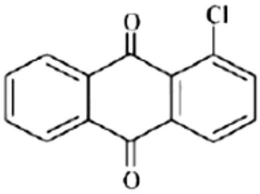
A.



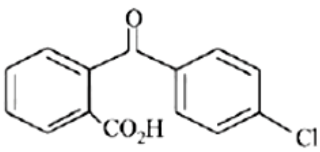
B.



C.

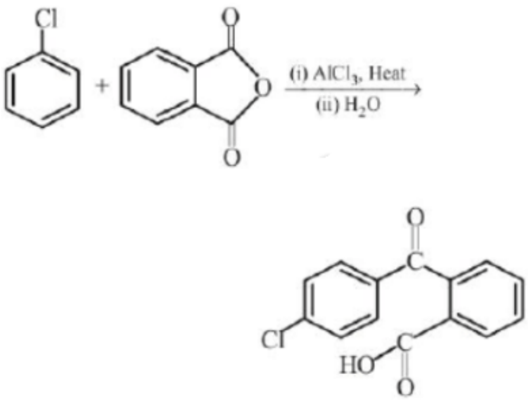


D.



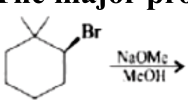
Answer: D

Solution:



Question111

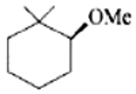
The major product of the following reaction is:



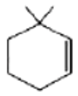
[2018]

Options:

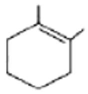
A.



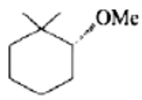
B.



C.



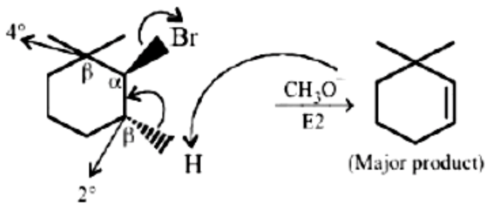
D.



Answer: B

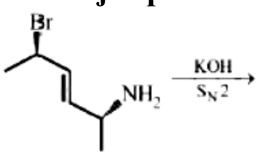
Solution:

CH_3O^- is a strong base and strong nucleophile, so favourable condition is $S_N2/E2$.
The given alkyl halide is 2° and β carbons are 4° and 2°, so sufficiently hindered, thus $E2$ dominates over S_N2



Question112

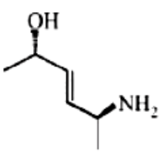
The major product of the following reaction is:



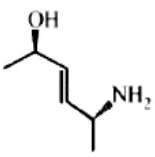
[Online April 16,2018]

Options:

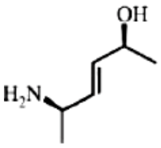
A.



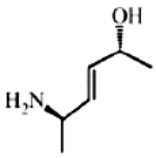
B.



C.

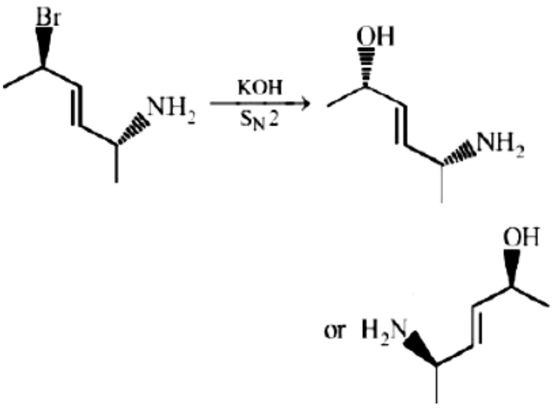


D.



Answer: C

Solution:



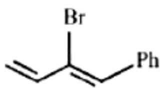
Inversion takes place at the carbon containing bromine atom.

Question113

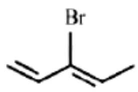
Which of the following will most readily give the dehydrohalogenation product? [Online April 15,2018(I)]

Options:

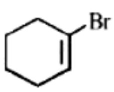
A.



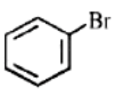
B.



C.



D.



Answer: A

Solution:

Here dehydrohalogenation goes by $\mathrm{E\,1cB}$ and most stable carbanion formation is favoured in (a).

Question114

The major product of the following reaction is:

$$\mathrm{CH_3\underset{\substack{|\\Br}}{C}HCH_2\underset{\substack{|\\Br}}{C}HCH_2CH_3\,Br} \xrightarrow[\text{heat}]{\mathrm{KOH,\,CH_3OH}}$$

[Online April 8,2017]

Options:

A. $\mathrm{CH_2=CHCH_2CH=CHCH_3}$

B. $\mathrm{CH_2=CHCH=CHCH_2CH_3}$

C. $\mathrm{CH_3CH=C=CHCH_2CH_3}$

D. $\mathrm{CH_3CH=CH-CH=CHCH_3}$

Answer: D

Solution:

Solution:

$$\mathrm{CH_3-\underset{\substack{|\\Br}}{CH}-CH_2-\underset{\substack{|\\Br}}{CH}-CH_2-CH_3} \xrightarrow[\mathrm{E2}]{\mathrm{KOH\,.\,CH_3OH/\Delta}} \mathrm{CH_3-CH=\underset{\text{(Saytzeff product)}}{CH-CH}=CH_2-CH_3}$$

Question115

The major product of the following reaction is $\mathrm{C_6H_5CH_2-\overset{\substack{CH_3\\|}}{\underset{\substack{|\\Br}}{C}}-CH_2-CH_3} \xrightarrow[\mathrm{C_2H_5OH}]{\mathrm{C_2H_5ONa}}$

[Online April 8,2017]

Options:

A. $\mathrm{C_6H_5CH_2-\overset{\substack{CH_3\\|}}{\underset{\substack{|\\OC_2H_5}}{C}}-CH_2-CH_3}$

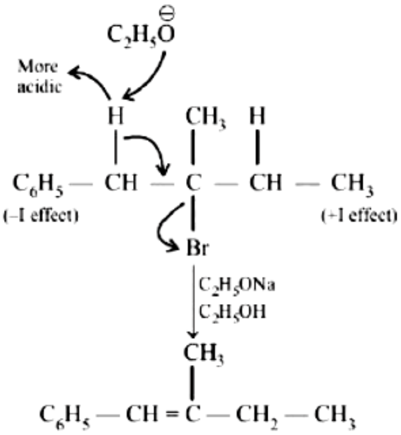
B. $\mathrm{C_6H_5CH-\overset{\substack{CH_3\\|}}{\underset{\substack{|\\CH_3}}{C}}-CH_2-CH_3}$

C. $\mathrm{C_6H_5CH=\underset{\substack{|\\CH_3}}{C}-CH_2-CH_3}$

D. $\mathrm{C_6H_5CH_2-\underset{\substack{|\\CH_2CH_3}}{C}=CH_2}$

Answer: C

Solution:



Question116

In the following reaction sequence:

$$\underset{(C_3H_6Cl_2)}{I} \xrightarrow{KOH(aq)} II \xrightarrow[(ii)H_2O/H^+]{(i)CH_3MgBr} III \xrightarrow{Anhy.\,ZnCl_2 + Conc.\,HCL} \text{given turbidity immediately}$$

The compound I is :
[Online April 9,2017]

Options:

A. $\underset{|\atop Cl}{CH_2} - \underset{|\atop Cl}{CH} - CH_3$

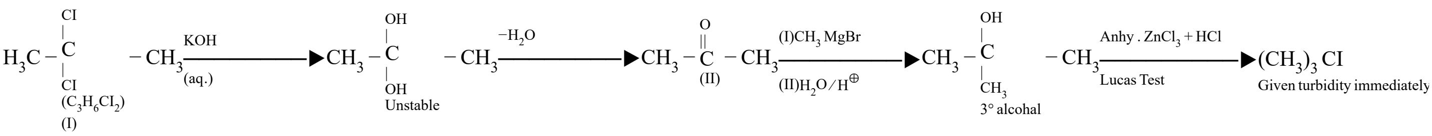
B. $\underset{|\atop Cl}{CH_2} - CH_2 - \underset{|\atop Cl}{CH_3}$

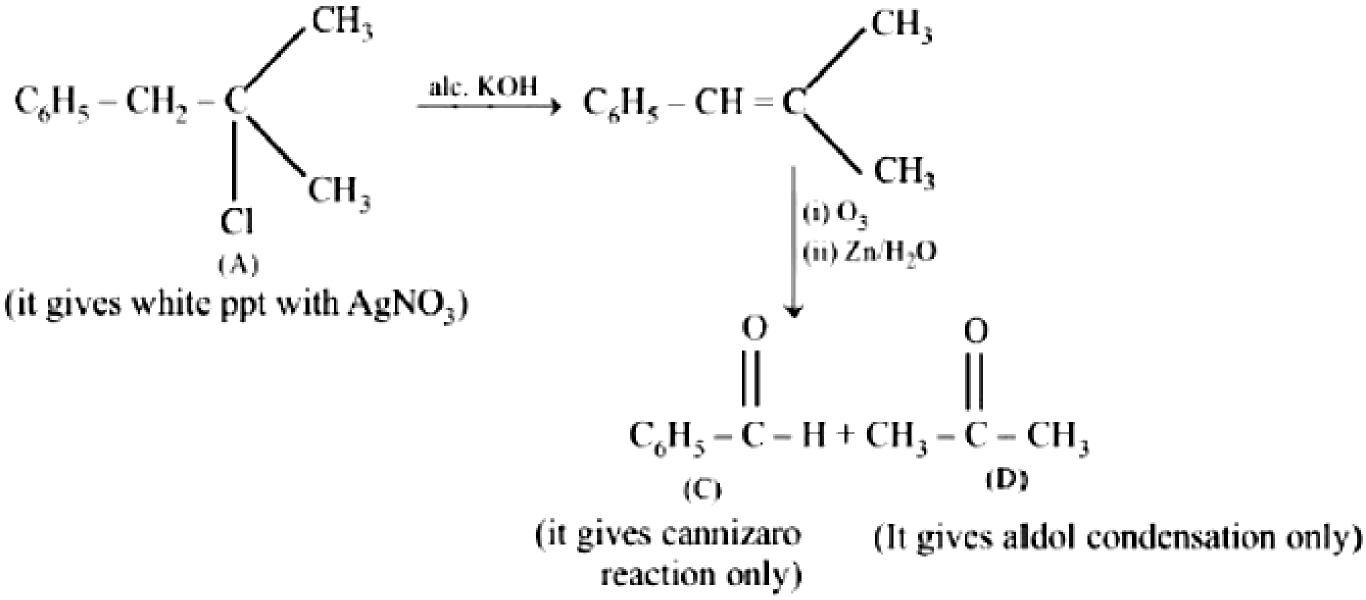
C. $CH - \underset{|\atop Cl}{CH} - CH_2 - CH_3$

D. $CH_3 - \overset{Cl\atop |}{\underset{|\atop Cl}{C}} - CH_3$

Answer: D

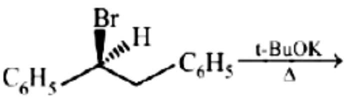
Solution:





Question117

The major product obtained in the following reaction is:



[2017]

Options:

A. (±)$C_6H_5CH(O^tBu)CH_2C_6H_5$

B. $C_6H_5CH = CHC_6H_5$

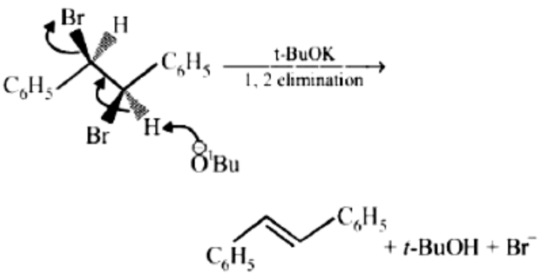
C. (+)$C_6H_5CH(O^tBu)CH_2C_6H_5$

D. (−)$C_6H_5CH(O^tBu)CH_2C_6H_5$

Answer: B

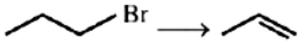
Solution:

Elimination reaction is highly favoured if (a) Bulkier base is used (b) Higher temperature is used Hence in given reaction biomolecular elimination reaction provides major product.



Question118

Which one of the following reagents is not suitable for the elimination reaction?



[Online April 10,2016]

Options:

A. NaI

B. NaOEt / EtOH

C. NaOH / H_2O

D. NaOH / H_2O − EtOH

Answer: A

Solution:

Alkyl chloride or bromide undergo substitution and get converted to an alkyl iodide on treatment with a solution of sodium iodide in acetone. e.g.

$CH_3CH_2CH_2Br + NaI \xrightarrow{\text{acetone}} CH_3CH_2CH_2I + NaBr$

This reaction is also known as Finkelstein Reaction.

Question119

The synthesis of alkyl fluorides is best accomplished by:
[2015]

Options:

A. Finkelstein reaction

B. Swarts reaction

C. Free radical fluorination

D. Sandmeyer's reaction

Answer: B

Solution:

Alkyl fluorides are more conveniently prepared by heating suitable chloro − or bromo-alkanes with organic fluorides such as AsF_3, SbF_3, CoF_2, AgF, Hg_2F_2 etc. This reaction is called Swarts reaction.

$CH_3Br + AgF \longrightarrow CH_3F + AgBr$

$2CH_3CH_2Cl + Hg_2F_2 \longrightarrow 2CH_3CH_2F + Hg_2Cl_2$

Question120

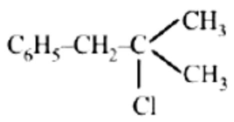
A compound A with molecular formula $C_{10}H_{13}Cl$ gives a white precipitate on adding silver nitrate solution. A on reacting with alcoholic KOH gives compound B as the main product. B on ozonolysis gives C and D. C gives Cannizaro reaction but not aldol condensation. D gives aldol condensation but not Cannizaro reaction. A is:
[Online April 10,2015]

Options:

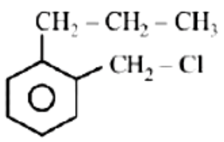
A. $C_6H_5 - CH_2 - CH_2 - CH_2 - CH_2 - Cl$

B. $C_6H_5-CH_2-CH_2-\underset{\substack{| \\ Cl}}{CH}-CH_3$

C.



D.



Answer: C

Solution:

Compound A reacts with alc.KOH to give compound B which on further ozonolysis gives C (does not contains $\alpha-H$ atom) and D (contains $\alpha-H$ atom). This reaction sequence can be achieved by compounds in option (a) and (c). Since compound A gives white ppt. with $AgNO_3$ preferable option will be (c) as tert alkyl reacts with $AgNO_3$ more quickly.

Question121

In S_N2 reactions, the correct order of reactivity for the following compounds: CH_3Cl, CH_3CH_2Cl, $(CH_3)_2CHCl$ and $(CH_3)_3CCl$ is:
[2014]

Options:

A. $CH_3Cl > (CH_3)_2CHCl > CH_3CH_2Cl > (CH_3)_3CCl$

B. $CH_3Cl > CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

C. $CH_3CH_2Cl > CH_3Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

D. $(CH_3)_2CHCl > CH_3CH_2Cl > CH_3Cl > (CH_3)_3CCl$

Answer: B

Solution:

Steric hindrance around the carbon atom having Cl will slow down the S_N2 reaction, hence lesser the hindrance, faster will be the reaction. So, the order of reactivity is $CH_3Cl > (CH_3)CH_2-Cl > (CH_3)_2CH-Cl > (CH_3)_3CCl$

Question122

For the compounds CH_3Cl, CH_3Br, CH_3I and CH_3F, the correct order of increasing C-halogen bond length is:
[Online April 9,2014]

Options:

A. $CH_3F < CH_3Cl < CH_3Br < CH_3I$

B. $CH_3F < CH_3Br < CH_3Cl < CH_3I$

C. $CH_3F < CH_3I < CH_3Br < CH_3Cl$

D. $CH_3Cl < CH_3Br < CH_3F < CH_3I$

Answer: A

Solution:

The correct order of increasing bond length is
$CH_3F < CH_3Cl < CH_3Br < CH_3I$

Question123

In a nucleophilic substitution reaction:
$$R-Br+Cl^- \xrightarrow{DMF} R-Cl+Br^-$$
which one of the following undergoes complete inversion of configuration? [Online April 9. 2014]

Options:

A. $C_6H_5CHC_6H_5Br$

B. $C_6H_5CH_2Br$

C. $C_6H_5CHCH_3Br$

D. $C_6H_5CCH_3C_6H_5Br$

Answer: C

Solution:

$C_6H_5CHCH_3Br$ being an optically active secondary alkyl bromide undergoes S_N2 nucleophilic substitution reaction. Hence it undergoes complete inversion of configuration.

Question124

The major organic compound formed by the reaction of 1, 1, 1 -trichloroethane with silver powder is: [2014]

Options:

A. Acetylene

B. Ethene

C. 2 - Butyne

D. 2 - Butene

Answer: C

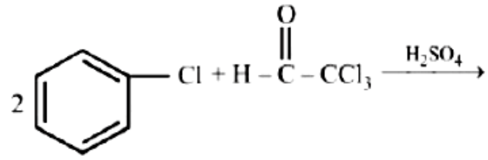
Solution:

$$2\,CI-\underset{\underset{CI}{|}}{\overset{\overset{CI}{|}}{C}}-CH_3+6\,Ag \longrightarrow \underset{2-\text{butyne}}{CH_3C}\ = CCH_3+6\,AgCI$$

1, 1, 1 − trichloroethane

Question125

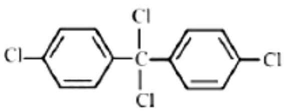
Chlorobenzne reacts with trichloro acetaldehyde in the presence of H_2SO_4



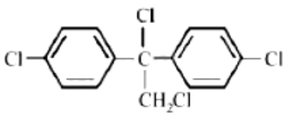
The major product formed is:
[Online April 11, 2014]

Options:

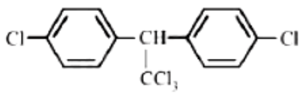
A.



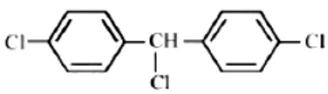
B.



C.



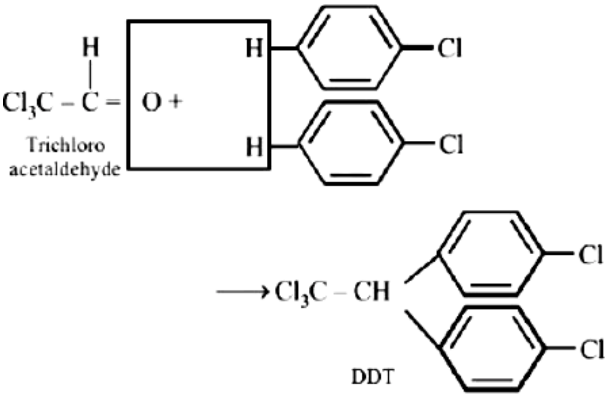
D.



Answer: C

Solution:

Chloral on reaction with chlorobenzene in the presence of a catalytic amount of sulphuric acid forms DDT (dichloro diphenyl trichloro ethane).



Question126

The major product formed when 1,1,1 -trichloro propane is treated with aqueous potassium hydroxide, is:
[Online April 19, 2014]

Options:

A. propyne

B. 1 -propanol

C. 2 -propanol

D. propionic acid

Answer: D

Solution:

$$\underset{1,\,1,\,1-\text{trichloro}-\text{propane}}{CI_3C-CH_2CH_3}+KOH\xrightarrow{\text{heat}}(OH)_3C-CH_2CH_3+3\,KCI\longrightarrow\underset{\text{propionic acid}}{CH_3CH_2\overset{\overset{O}{\|}}{C}OH}$$

Question127

The order of reactivity of the given haloalkanes towards nucleophile is :
[Online April 23,2013]

Options:

A. $RI > RBr > K\,Cl$

B. $RCl > RBr > RI$

C. $RBr > RCl > RI$

D. $RBr > RI > RCl$

Answer: A

Solution:

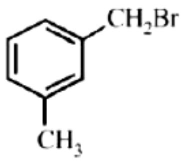
For a given alkyl group, the order of reactivity is
, $\dfrac{R-I>R-Br>R-Cl>R-F}{\text{increasing bond energy}}$ decreasing halogen reactivity.
This order depends on the carbon-halogen bond energy; the carbon-fluorine bond energy is maximum and thus fluorides are least reactive while carbon iodine bond energy is minimum hence iodides are most reactive.

Question128

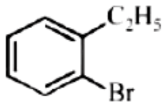
Compound (A), C_8H_9Br, gives a yellow precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_-$. (B) easily forms anhydride on heating. Identify the compound (A).
[2013]

Options:

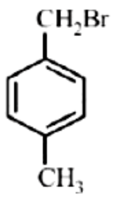
A.



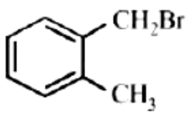
B.



C.

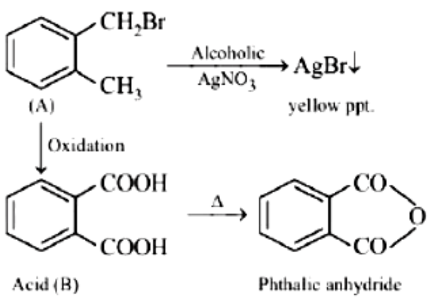


D.



Answer: D

Solution:



Question129

The Wurtz-Fittig reaction involves condensation of :
[Online April 22. 2013]

Options:

A. two molecules of aryl halides

B. one molecule of each of aryl-halide and alkyl-halide.

C. one molecule of each of aryl-halide and phenol.

D. two molecules of aralkyl-halides.

Answer: B

Solution:

Reaction between alkyl halides, aryl halides and sodium in presence of dry ether to give substituted aromatic compounds is known as Wurtz fitting reaction $C_6H_5Cl + 2Na + ClCH_3 \longrightarrow \underset{Toluene}{C_6H_5 \quad CH_3} + 2NaCl$

Question130

Alkyl halides react with dialkyl copper reagents to give [2005]

Options:

A. alkenyl halides

B. alkanes

C. alkyl copper halides

D. alkenes

Answer: A

Solution:

In Corey House synthesis of alkanes alkyl halides react with lithium dialkyl cuprate
$R'X + LiR_2Cu \longrightarrow R' - R + RCu + LiX$

Question131

Aryl fluoride may be prepared from arene diazonium chloride using: [Online April 9. 2013]

Options:

A. HBF_4/Δ

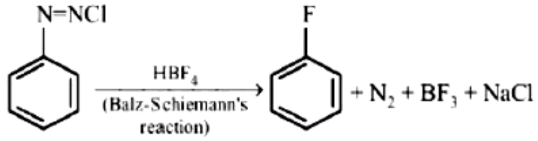
B. $HBF_4/NaNO_2Cu, \Delta$

C. CuF/HF

D. Cu/HF

Answer: A

Solution:



Question132

Elimination of bromine from 2 -bromobutane results in the formation of -
[2005]

Options:

A. Predominantly 2 -butyne

B. Predominantly 1-butene

C. Predominantly 2 -butene

D. Equimolar mixture of 1 and 2 -butene

Answer: C

Solution:

-+9 $CH_3-\underset{|}{\overset{Br}{C}}H-CH_2-CH_3 \xrightarrow{Alc.\,KOH} CH_3-CH=CH-CH_3+HBr$

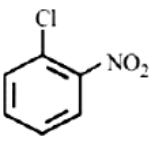
The formation of 2 -butene is in accordance to Saytzeff's rule (more substituted alkene is formed).

Question133

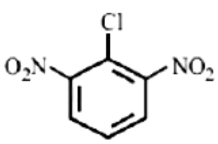
A major component of Borsch reagent is obtained by reacting hydrazine hydrate with which of the following?
[Online April 22. 2013]

Options:

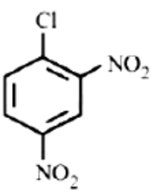
A.



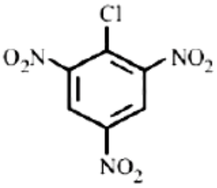
B.



C.



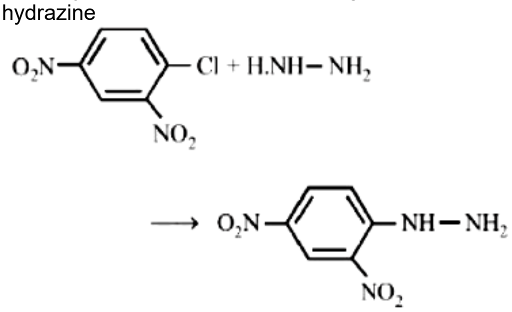
D.



Answer: C

Solution:

The major component of Borsch reagent is 2,4 dinitrophenyl hydrazine which can be obtained by reaction of 2,4 -dinitrochloro benzene and hydrazine



2,4 -Dinitrophenyl hydrazine

Question134

How many chiral compounds are possible on monochlorination of 2 - methyl butane? [2012]

Options:

A. 8

B. 2

C. 4

D. 6

Answer: D

Solution:

$$\overset{\overset{Cl}{|}}{C}H_2-\overset{*}{C}H(CH_3)-CH_2-CH_3 \quad (R+S)$$

$$CH_3-CH(CH_3)-\overset{\overset{Cl}{|}}{\underset{*}{C}}H-CH_3 \quad (R+S)$$

Four monochloro derivatives are chiral.

Question135

Phenyl magnesium bromide reacts with methanol to give [2005]

Options:

A. a mixture of toluene and M g(OH)Br

B. a mixture of phenol and M g(M e)Br

C. a mixture of anisole and $Mg(OH)Br$

D. a mixture of benzene and $Mg(OMe)Br$

Answer: D

Solution:

$$CH_3OH + C_6H_5MgBr \longrightarrow CH_3OMgBr + C_6H_6$$

Question136

$C_2H_5Br \xrightarrow{AgCN} X \xrightarrow[Zn-Hg/HCl]{Reduction} Y$, Here Y is
[Online May 7, 2012]

Options:

A. Ethyl methyl amine

B. n -propylamine

C. Isopropylamine

D. Ethylamine

Answer: B

Solution:

$$C_2H_5Br \xrightarrow{AgCN} \underset{(X)}{C_2H_5NC} + AgBr$$

$$\underset{(X)}{C_2H_5NC} \xrightarrow[Zn-Hg/HCl]{Reduction} \underset{(Y)\ \text{ethyl methyl amine}}{C_2H_5NHCH_3}$$

Question137

The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid, is
[2004]

Options:

A. freon

B. DDT

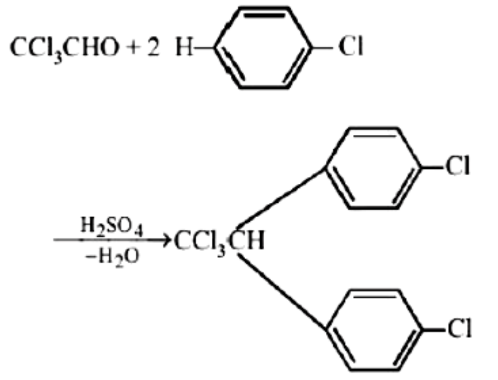
C. gammexene

D. hexachlorocthane

Answer: B

Solution:

DDT is prepared by heating chlorbenzene and chloral with concentrated sulphuric acid



1,1,1-Trichloro-2,2 bis (p-chlorophenyl) ethane or DDT

Question138

Which of the following statements is wrong?
[Online May 12, 2012]

Options:

A. Ethyl chloride on reduction with Zn -Cu couple and alcohol gives ethane.

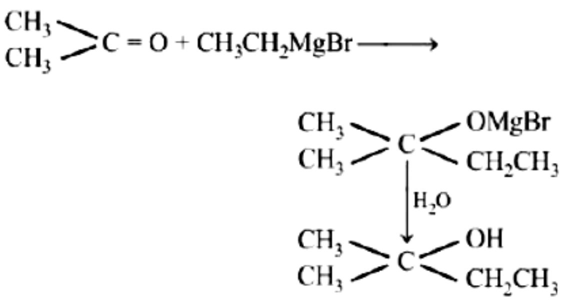
B. The reaction of methyl magnesium bromide with acetone gives butanol- 2 .

C. Alkyl halides follow the following reactivity sequence on reaction with alkenes. $R-I > R-Br > R-Cl > R-I$

D. $C_2H_4Cl_2$ may exist in two isomeric forms

Answer: D

Solution:



2- Methyl butanol-2
(3° alcohol)

Question139

Bottles containing C_6H_5I and $C_6H_5CH_2I$ lost their original labels. Theywere labelled A and B for testing. A and B were separately taken in test tubes and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO_3 and then some $AgNO_3$ solution was added. Substance B gave a yellow precipitate. Which one of the following statements is true for this experiment?
[2003]

Options:

A. A is $C_6H_5CH_2I$

B. B is C_6H_5I

C. Addition of HNO_3 was unnecessary

D. A is C_6H_5I

Answer: D

Solution:

$$\underset{(A)}{C_6H_5I} \xrightarrow{NaOH} C_6H_6ONa \xrightarrow{HNO_3/H^+}$$

$$C_6H_5OH \xrightarrow{AgNO_3} \text{No yellow ppt.}$$

$$\underset{(B)}{C_6H_5CH_2I} \xrightarrow{NaOH} C_6H_5CH_2ONa \xrightarrow{HNO_3/H^+} C_6H_5CH_2OH \xrightarrow{AgNO_3} \text{yellow ppt.}$$

Since benzyl iodide gives yellow ppt. hence this is compound B and A is phenyl iodide (C_6H_5I).

Question140

The reaction:

$(CH_3)_3C-Br \xrightarrow{H_2O} (CH_3)_3-C-OH$ [2002]

Options:

A. elimination reaction

B. substitution reaction

C. free radical reaction

D. displacement reaction.

Answer: D

Solution:

The hydrolysis of t -butyl bromide is an example of S_N1 reaction.

Question141

Among the following, the molecule with the lowest dipole moment is [Online May 19,2012]

Options:

A. $CHCl_3$

B. CH_3Cl

C. CH_2Cl_2

D. CCl_4

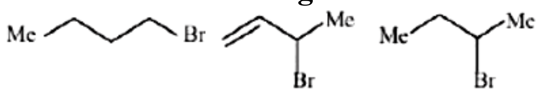
Answer: D

Solution:

CCl_4 is a nonpolar moleculas and it has symmetrical tetrahedral structure. Although each of the c -cl bond is polar but the resultant of all there dipole moments is zero.

Question142

Consider the following bromides :



(A) (B) (C)

The correct order of S_N I reactivity is
[2010]

Options:

A. B > C > A

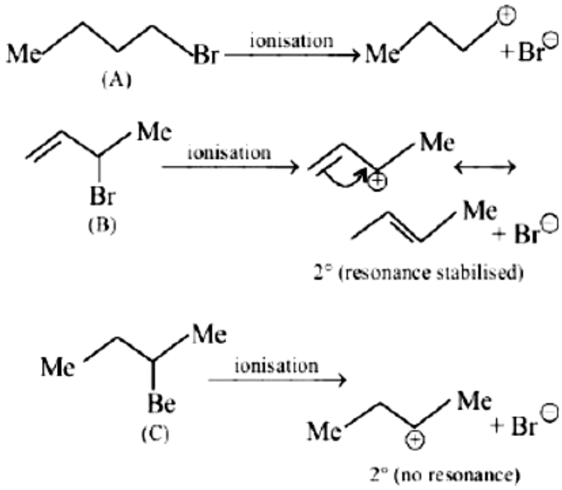
B. B > A > C

C. C > B > A

D. A > B > C

Answer: A

Solution:



Since S_N1 reactions involve the formation of carbocation as intermediate in the rate determining step, more is the stability of carbocation higher will be reactivity of alkyl halides towards S_N1 route. Now we know that stability of carbocations follows the order : $3^\circ > 2^\circ > 1^\circ$, so$S_N1$ reactivity should also follow the same order.
$3^\circ > 2^\circ > 1^\circ >$ Methyl (S_N1 reactivity)

Question143

The organic chloro compound, which shows complete sterochemical inversion during a S_N2 reaction, is
[2008]

Options:

A. $(C_2H_5)_2CHCl$

B. $(CH_3)_3CCl$

C. $(CH_3)_2CH\,Cl$

D. CH_3Cl

Answer: D

Solution:

S_N2 reaction is favoured by small groups on the carbon atom attached to halogen. So, the order of reactivity is
$CH_3Cl > (CH_3)_2CH\,Cl > (CH_3)_3CCl > (C_2H_5)_2CH\,Cl$

Question144

Which of the following is the correct order of decreasing S_N2 reactivity? [2007]

Options:

A. $R_2CH\,X > R_3CX > RCH_2X$

B. $RCH_2X > R_3CX > R_2CH\,X$

C. $RCH_2X > R_2CH\,X > R_3CX$

D. $R_3CX > R_2CH\,X > RCH_2X$ (X is a halogen)

Answer: C

Solution:

In S_N2 mechanism transition state is pentavelent. For bulky alkyl group it will havesterical hinderance and smaller alkyl group will favour the S_N2 mechanism. So the decreasing order of reactivity of alkyl halide towards S_N2 mechanism is
$RCH_2X > R_2CH\,X > R_3CX$

Question145

Reaction of trans 2 -phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces [2006]

Options:

A. 1-phenylcyclopentene

B. 3-phenylcyclopentene

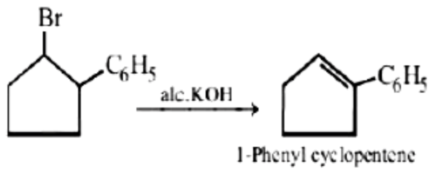
C. 2 -phenyleyclopentene

D. 4 -phenylcyclopentene

Answer: A

Solution:

The reaction is dehydrohalogenation



Question146

Fluorobenzene (C_6H_5F) can be synthesized in the laboratory
[2006]

Options:

A. by direct fluorination of benzene with F_2 gas

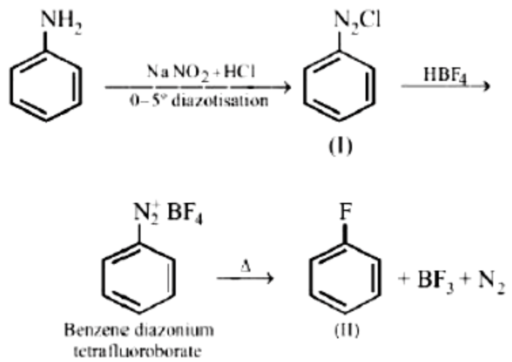
B. by reacting bromobenzene with NaF solution

C. by heating phenol with HF and KF

D. from aniline by diazotisation followed by heating the diazonium salt with HBF_4

Answer: D

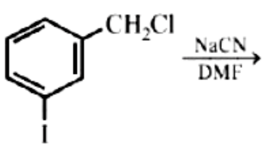
Solution:



Conversion of (I) to (II) is known as Balz-schilmann reaction.

Question147

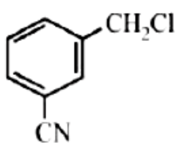
The structure of the major product formed in the following reaction is



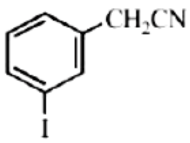
[2006]

Options:

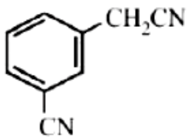
A.



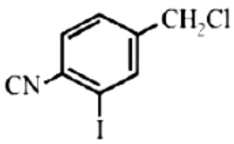
B.



C.

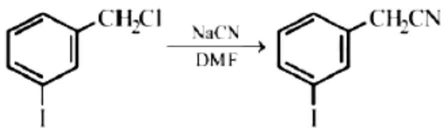


D.



Answer: B

Solution:



Nuclear substitution will not take place.

Question148

Tertiary alkyl halides are practically inert to substitution by S_N2 mechanism because of [2005]

Options:

A. steric hindrance

B. inductive effect

C. instability

D. insolubility

Answer: A

Solution:

Due to steric hindrance tertiary alkyl halides do not react by S_N2 mechanism, they react by S_N1 mechanism. S_N2 mechanism is followed in case of primary and secondary alkyl halides.
